

PHYS 2603



Physical Principles of Medical Imaging

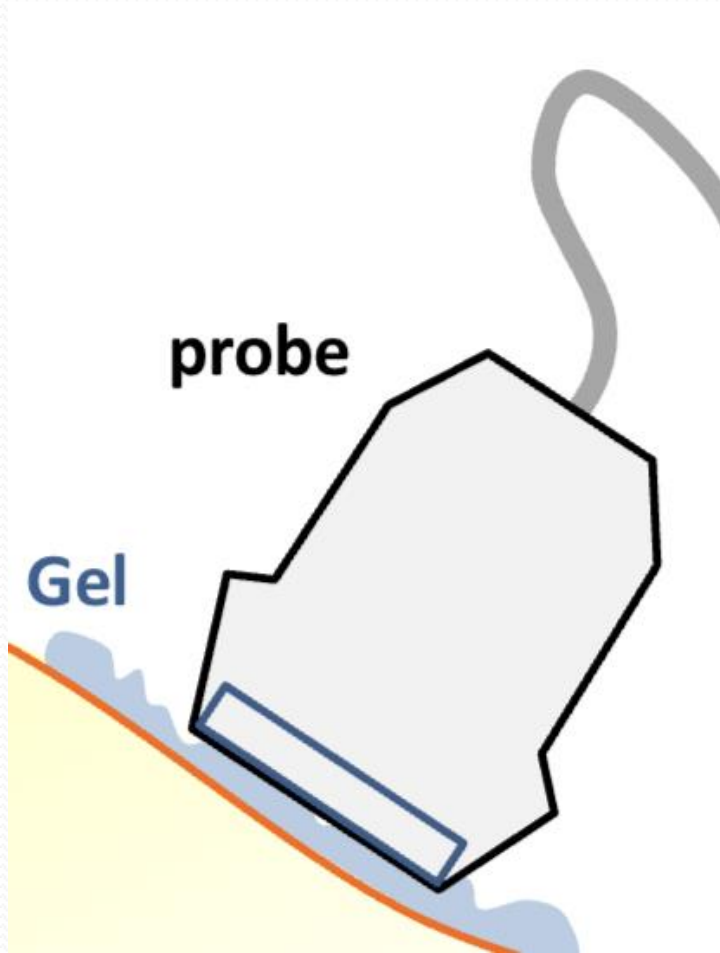
Course Review

Spring 2026



Ultrasound

Ultrasound Transducer



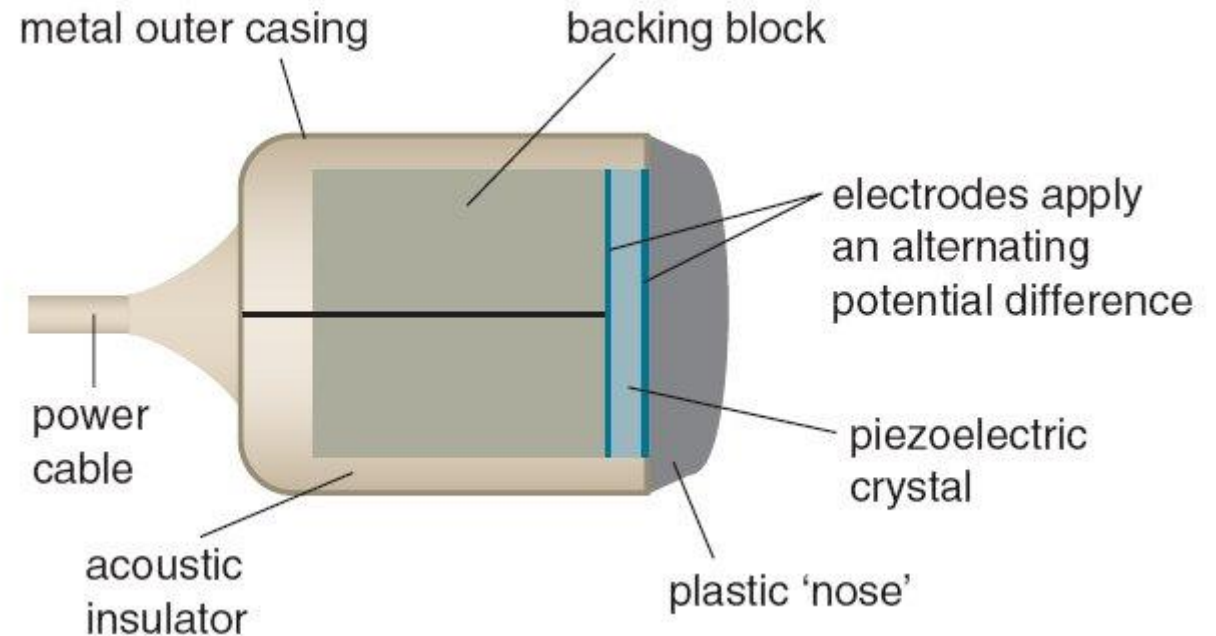
Generates ultrasound pulses

Transmits ultrasound inside the body

Detects and **converts** echoes into electronic signals

Piezoelectric Transducer

A device that can **convert alternating voltage into mechanical vibrations** (generating sound) and can also **convert mechanical vibrations into alternating voltage** (detecting echoes)



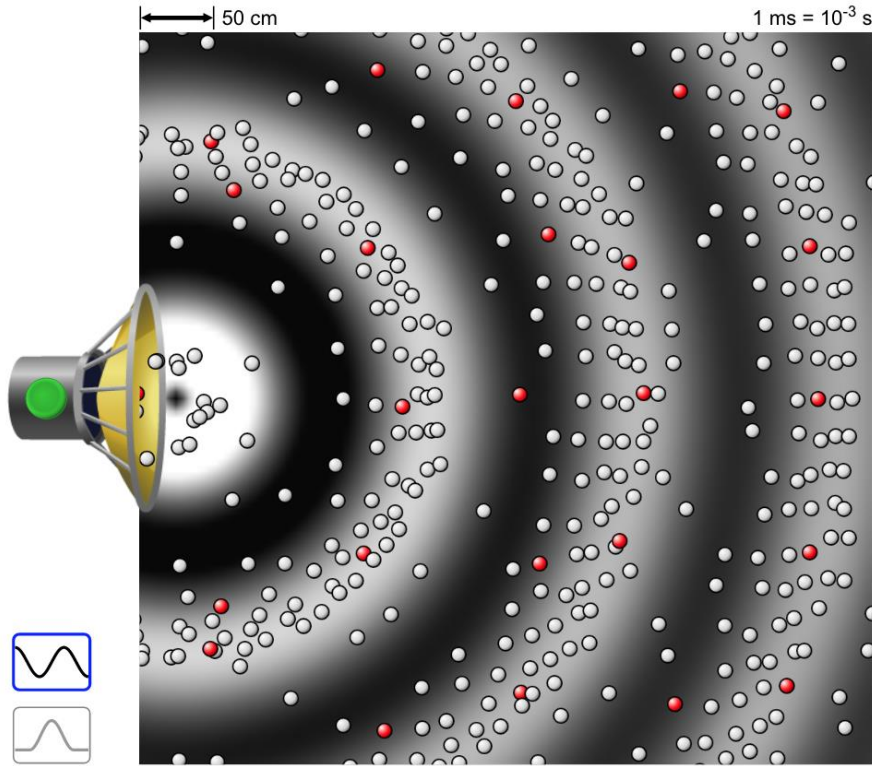
Sound Waves

In physics, the **sound** is defined as **collective vibrations** of molecules in gases, liquids and solids.

Sound vibrations occur when molecules are **squeezed** closer together temporarily forming a region of **higher pressure and density (compression)** surrounded by a low-density area (**rarefaction**).

Because of the collisions, the molecules tend to move into the low-density region **increasing** the **density** in the neighboring regions and leaving behind the **rarefied region**.

Sound Wave

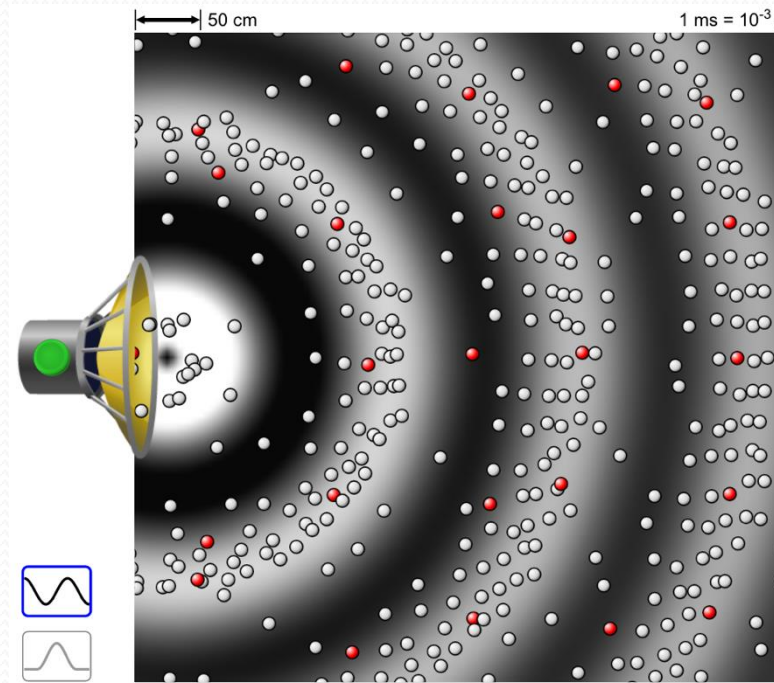


The **process** of pressure and density variations **spreading** from one region to another is called the **sound wave**

The wave **crests** correspond to the **maximum** values and wave **troughs** to the **minimum** values of pressure

Sound waves **propagate** (travel) as pressure at each point varies between the maximum and minimum value.

Sound Waves are Longitudinal Waves



The **directions along** which molecules are squeezed are the same in which the sound wave travels away from the source.

Therefore, the sound waves are **longitudinal waves**.

Period and Frequency of Vibrations

Vibrations are characterized by a period and frequency

Period:

The duration of one cycle of motion (one full swing)

$$T - \text{measured in seconds} \quad f = \frac{1}{T}$$

Frequency:

The number of cycles in one second (the rate of repetition)

$$f - \text{measured in Hertz} \quad (1 \text{ Hz} = \frac{1}{\text{sec}})$$

Frequency of sound. Ultrasound

Infrasound:	less than 20 Hz
Audible sound:	20 Hz – 20,000 Hz (20 kHz)
Ultrasound:	more than 20,000 Hz (20 kHz)
Diagnostic ultrasound:	2,000,000 Hz (2 MHz) - 15,000,000 Hz (15 MHz)

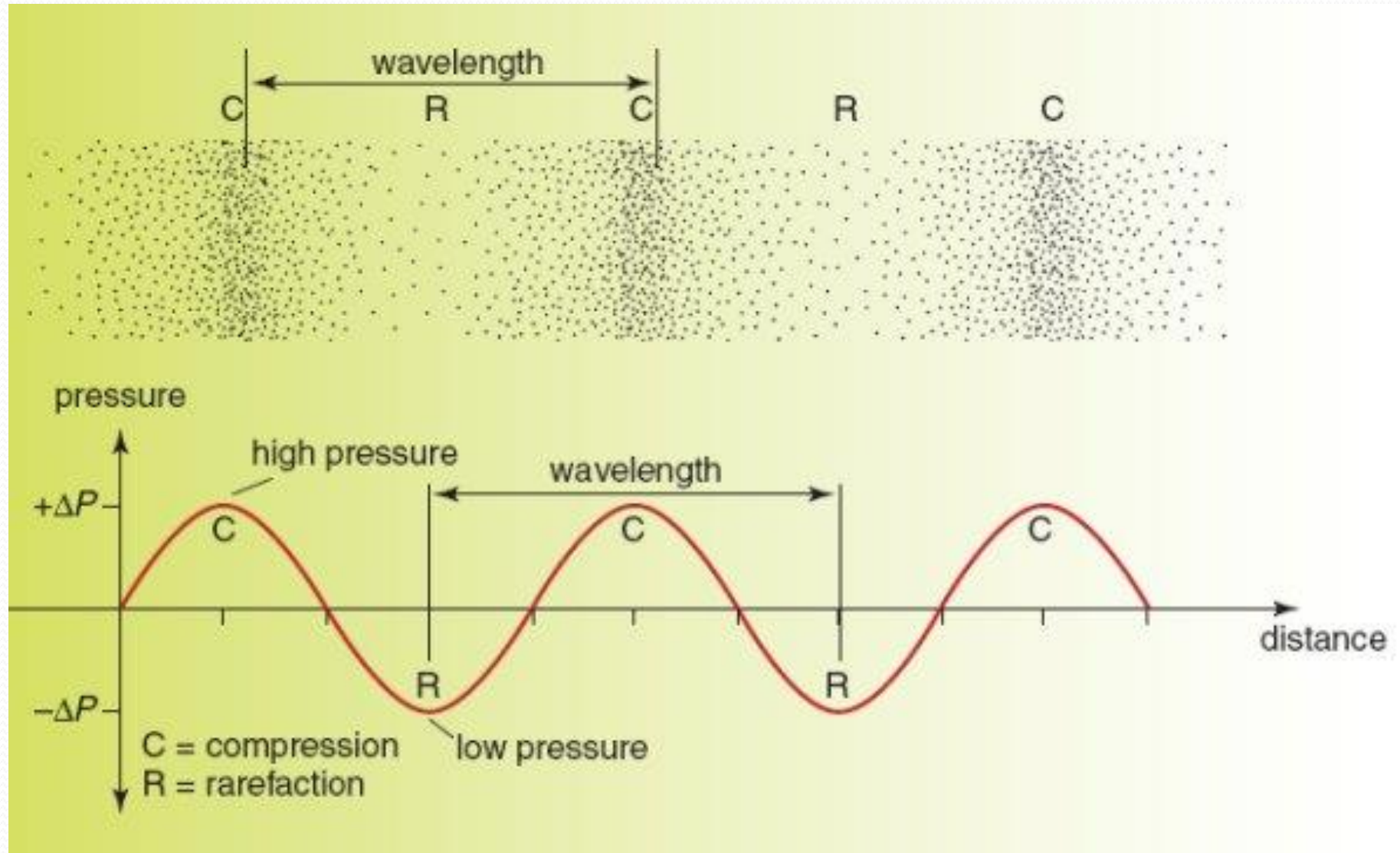
Wavelength

λ

(lambda)

The **spatial pattern** of pressure in a sound wave is characterized by **the wavelength** – the distance between the two nearest crests (high pressure) or two nearest troughs (low pressure)

Wavelength λ – the distance between crests



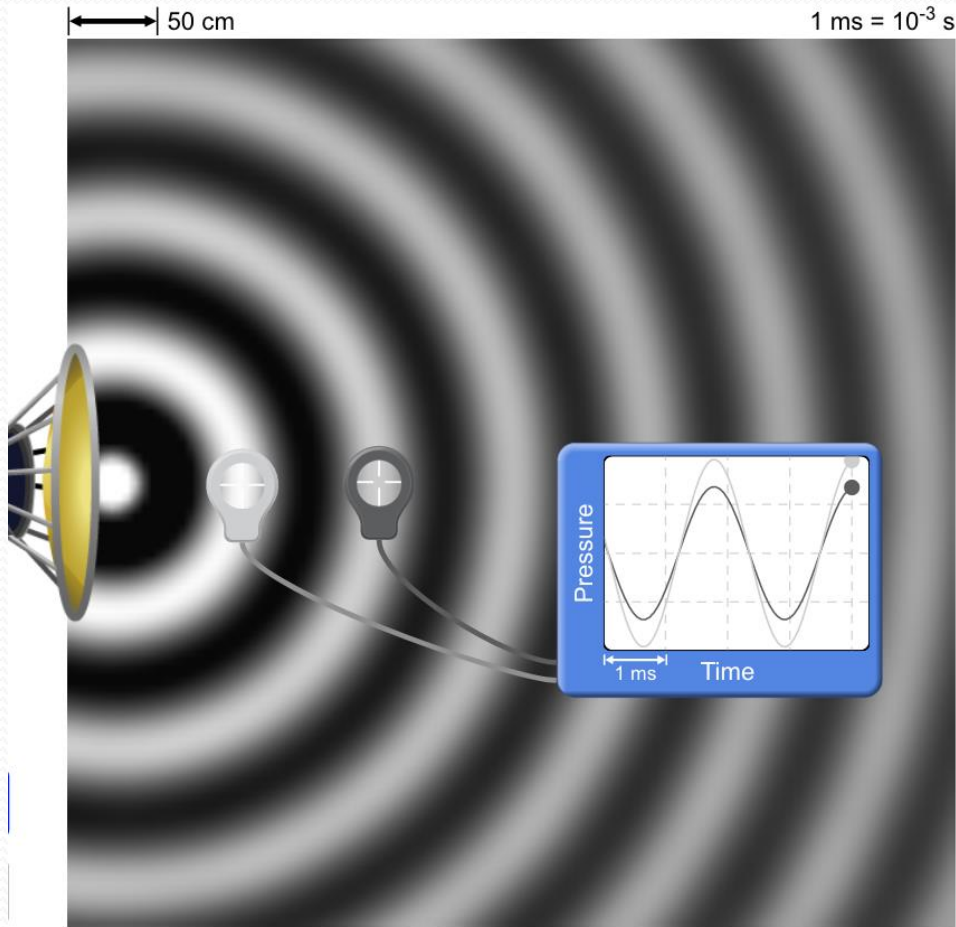
Speed of Sound v

Wave crests propagate (travel) in medium with a certain speed referred to as **the speed of sound**:

v

In time equal to **one period** a crest travels a distance equal to **one wavelength**:

$$v = \frac{\text{distance}}{\text{time}} = \frac{\lambda}{T} = f\lambda$$



Speed of Sound v

The values of the speed of sound are determined by the type of medium (tissue)

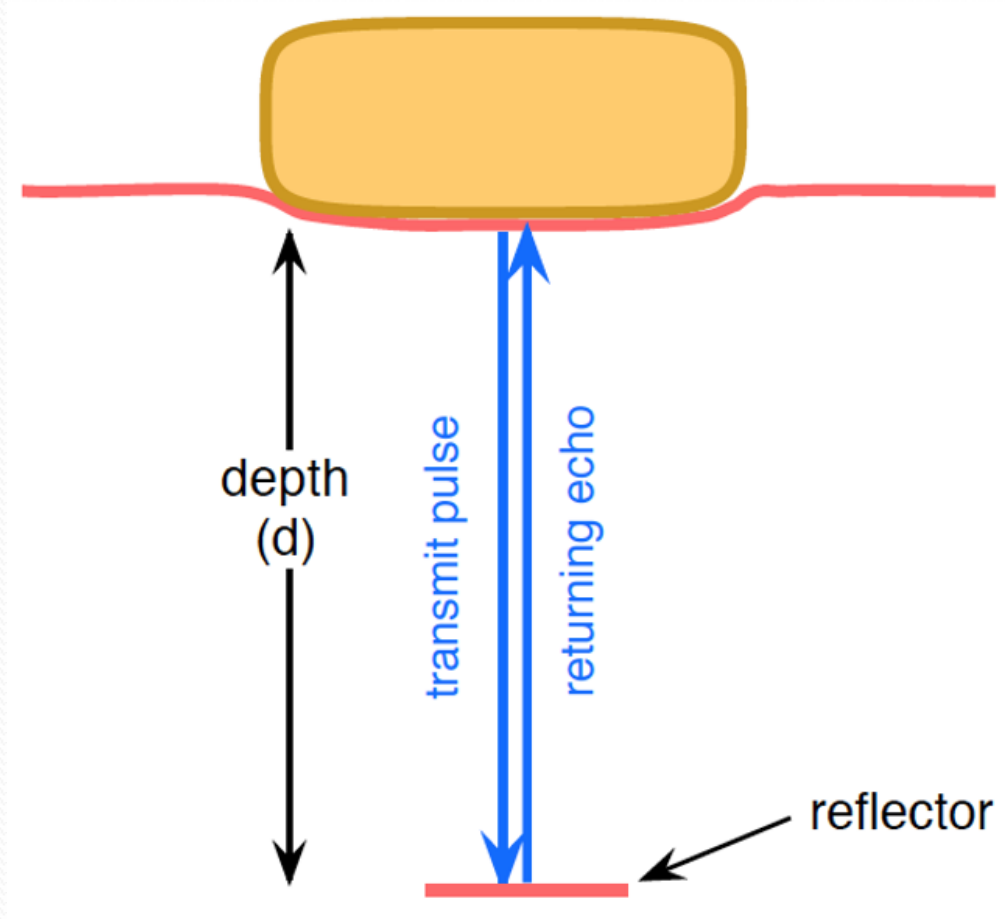
Materials / Tissues	Speed of sound m/s	Speed of sound $mm/\mu s$
Air	344	0.344
Acrylic (polyacrylate) plastic	2690	2.69
Water	1500	1.50
Soft tissue (average value)	1540	1.54
Muscle	1570	1.57
Fat	1440	1.44
Bone	3500	3.50

Echoes

Echoes are ultrasound pulses **reflected** from an **interface** separating two different tissues.

Medical ultrasound imaging is based on detecting echoes from the interface between the tissues inside the body.

Pulse-Echo Principle

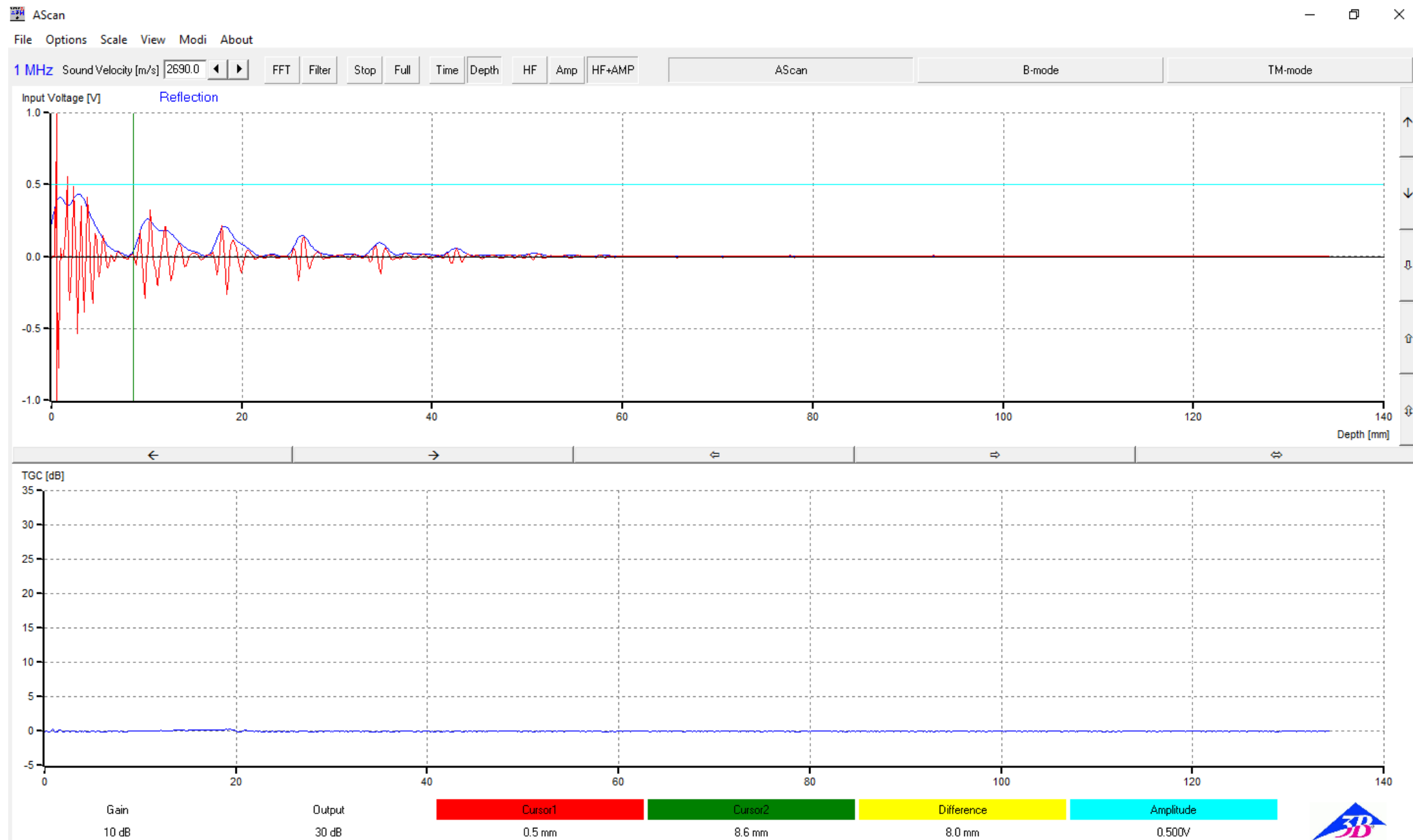


$$d = \frac{vt}{2}$$

d — depth to the interface

t — time between **emission** of a pulse and **arrival** of an **echo**

v — speed of sound in a tissue



Energy Carried by Sound

Ultrasound waves carry energy as they travel in tissues

Energy is determined by the amplitude of the wave
- the maximum pressure (the “height” of the crest)

The **intensity** of an ultrasound signal increases as the **square of the amplitude** to the signal

$$I \sim A^2$$

Acoustic Impedance

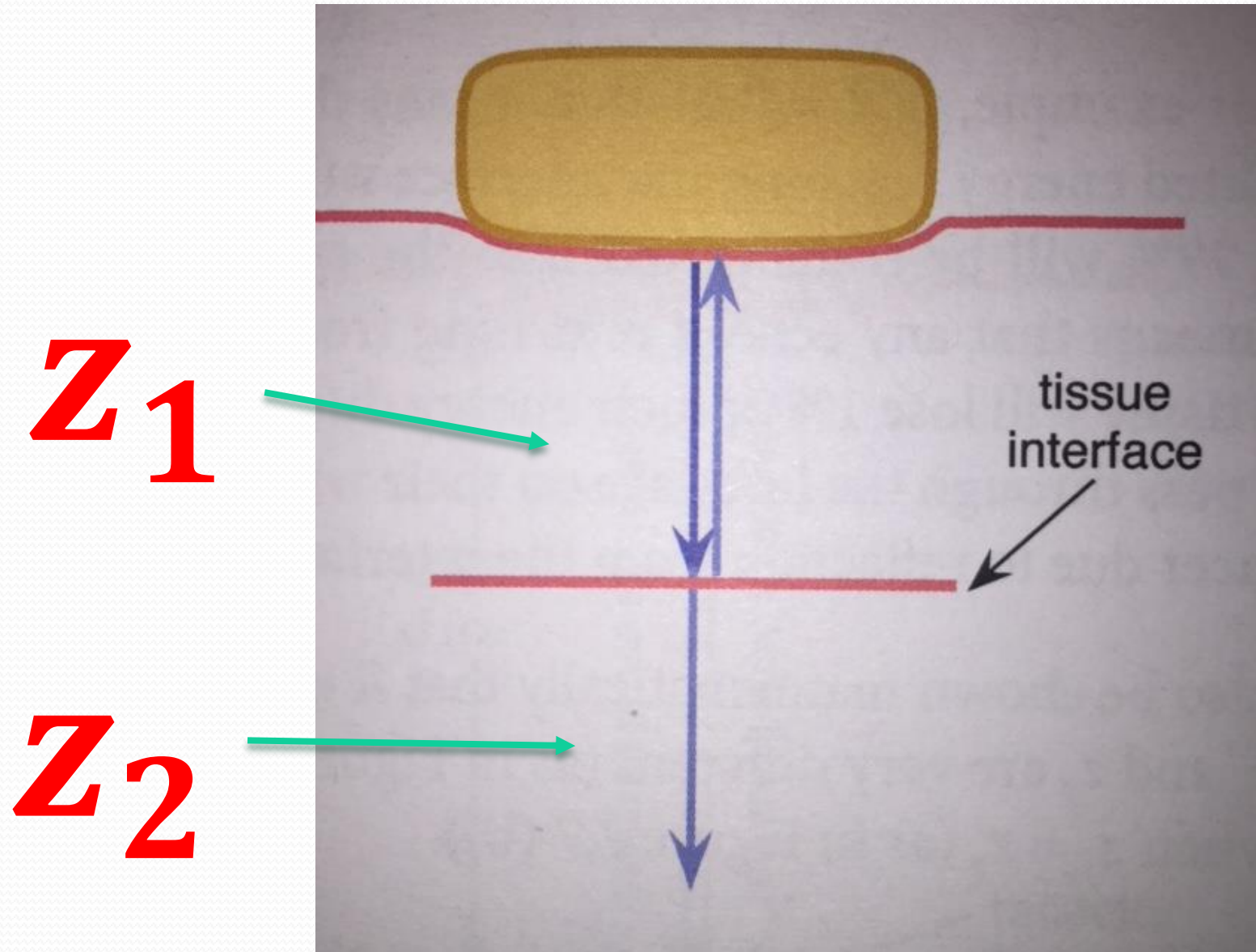
$$Z = \rho v_s$$

ρ	Density of tissue
v_s	Speed of sound

Acoustic Impedance

Materials / Tissues	Speed of sound v_s mm/ μ s	Acoustic Impedance z mg/(mm ² · μ s)
Air	0.344	0.0004
Acrylic (polyacrylate) plastic	2.69	3.20
Water	1.50	1.50
Muscle	1.57	1.65
Fat	1.44	1.30
Bone	3.50	7.80
Soft tissue (average value)	1.54	1.60

Intensity of Echoes



Reflection Coefficient

$$R = \frac{\text{Reflected Intensity}}{\text{Incident Intensity}}$$

$$R = \frac{I_{\text{refl}}}{I_0} = \frac{(z_2 - z_1)^2}{(z_2 + z_1)^2}$$

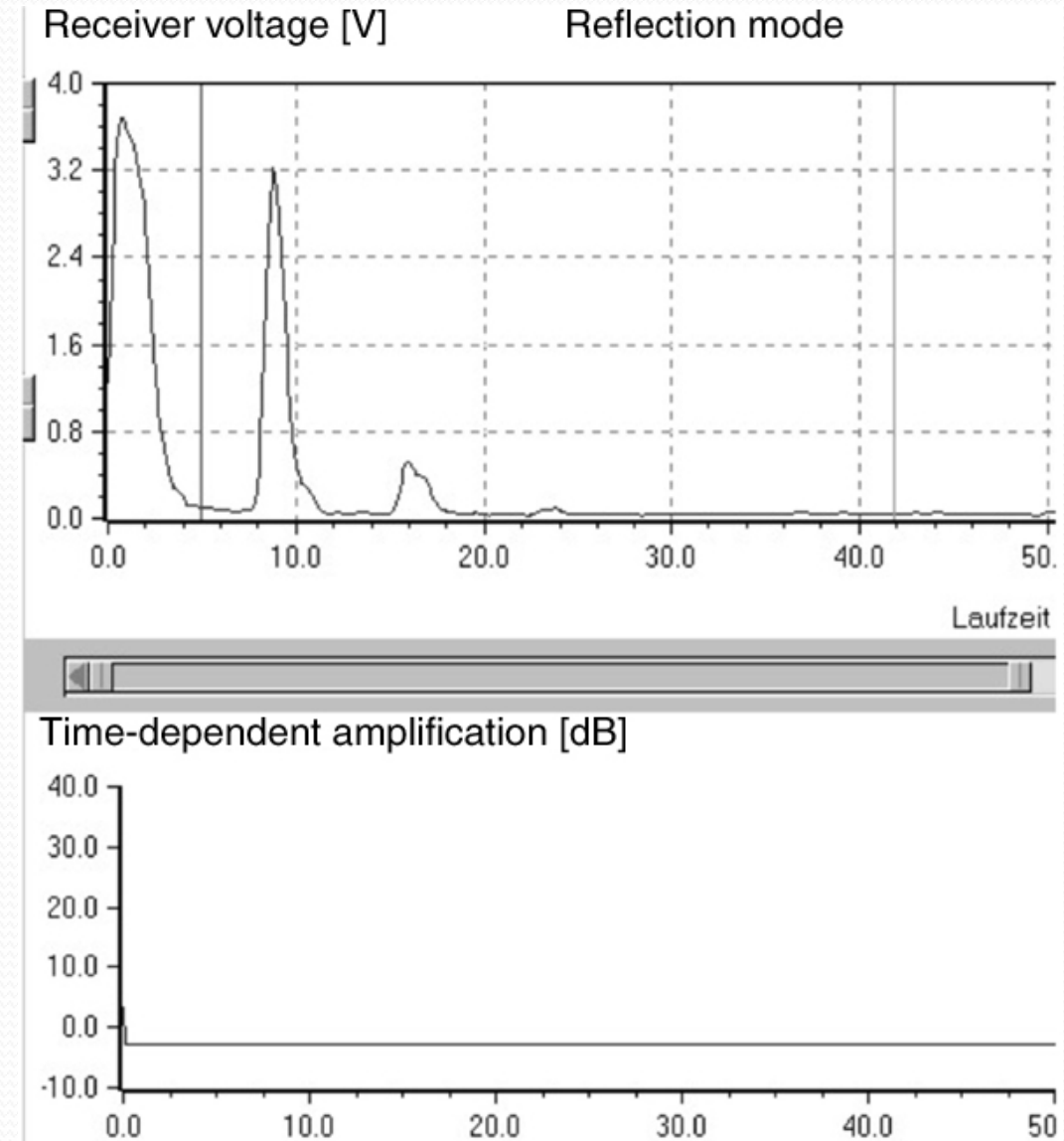
I_{refl} - Reflected Intensity (Echo Intensity)

I_0 - Incident Intensity (Emitted Pulse)

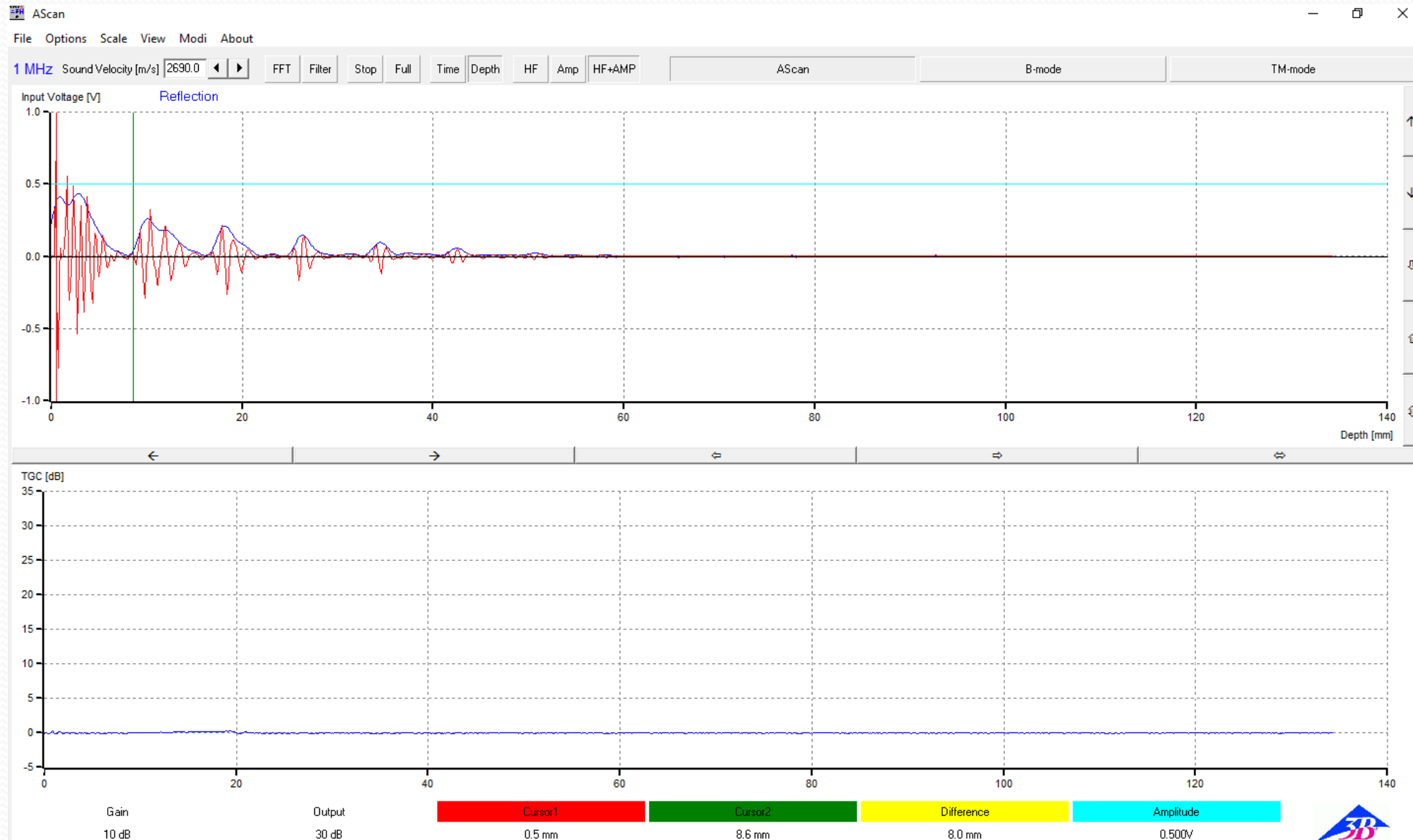
Ultrasound Absorption

The amplitude and intensity of echoes decreases with depth of travel

Energy carried by ultrasound is transferred to the molecules in tissues in the form of heat



Attenuation of Echoes: A-scans



Exponential Decrease of Intensity

$$I = I_0 e^{-x/D}$$

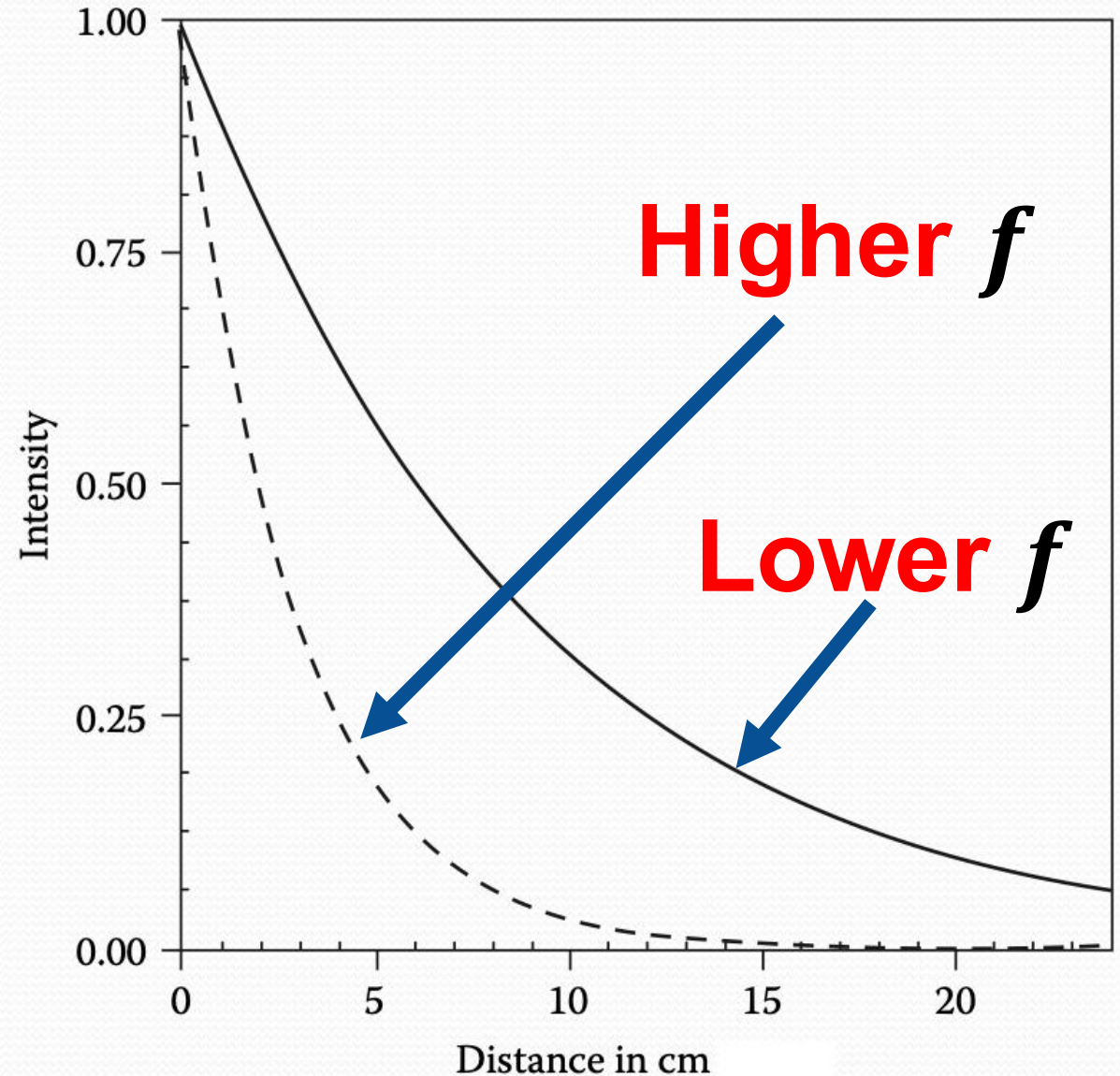
I_0 — Initial intensity of an ultrasound pulse

x — depth through which the pulse travels in tissue

D — a quantity (in units of length) describing the degree of attenuation

Ultrasound Attenuation and Frequency

Higher frequency ultrasound signals experience stronger absorption



B-Scans

Two dimensional maps of the echoes as function of the **position** of the transducer and the **depth** of the interface.

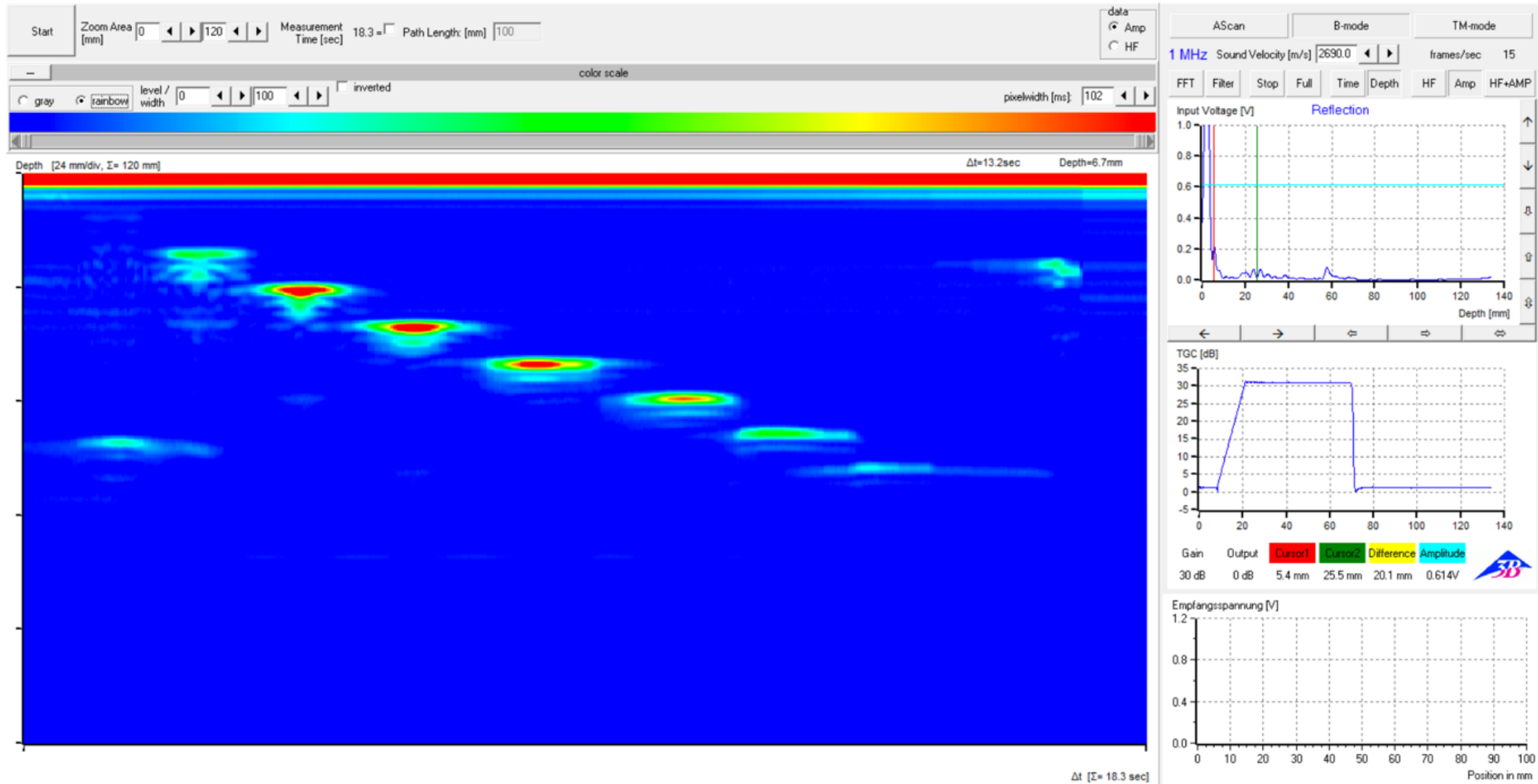
The **brightness** of a region in an image is determined by the **intensity** (**square of the voltage amplitude**) of the corresponding echo

Image Contrast

Image contrast is the difference in gray scale between two neighboring regions of tissue.

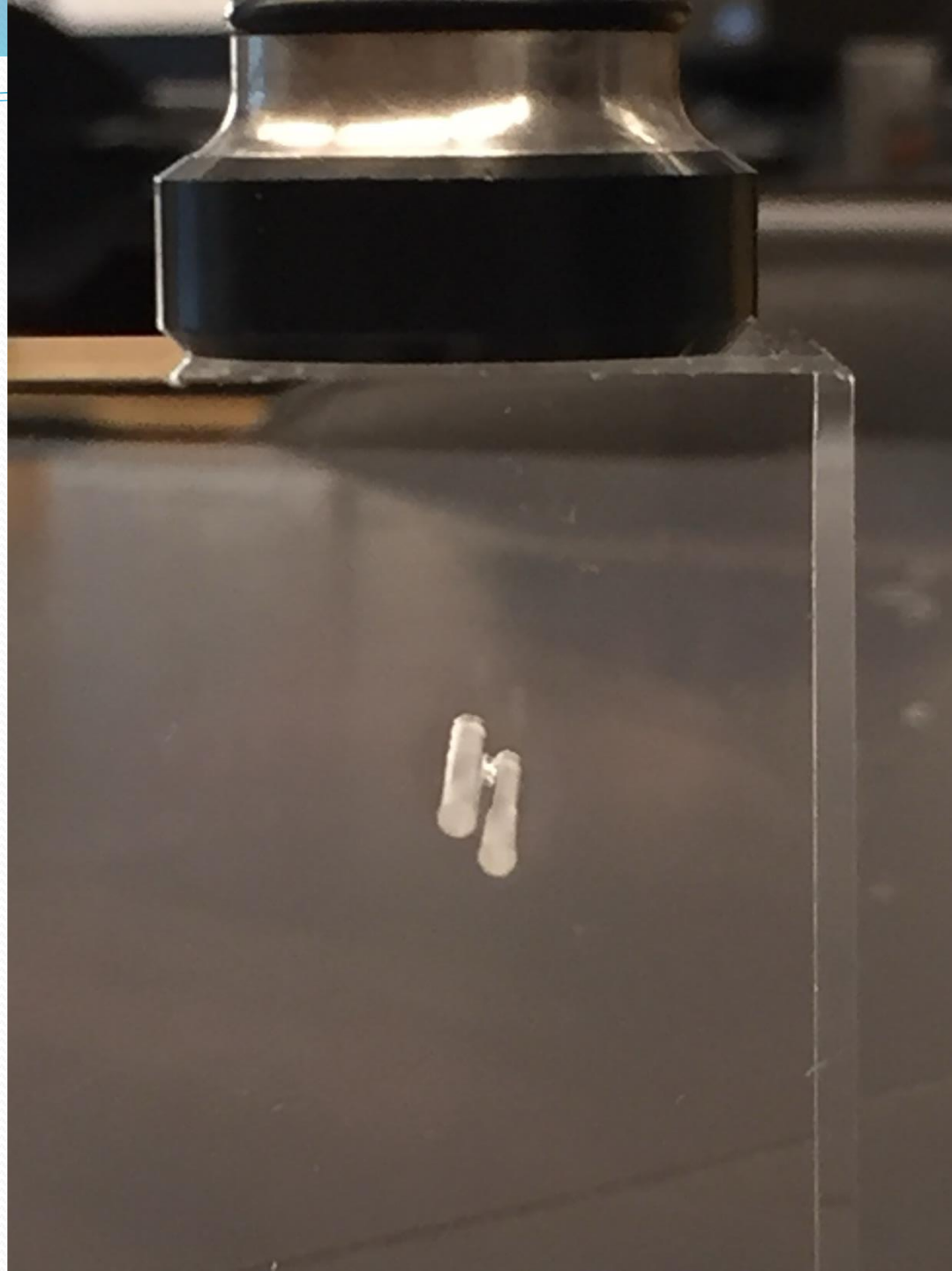
The image contrast depends on the difference in the acoustic impedance of the neighboring tissues.

Image Contrast: B-scan



Spatial Resolution

The **minimal distance** separating two adjacent points with the same reflection coefficient which are displayed in the image as two separate regions



Spatial Resolution

To **increase** spatial resolution, we need **increase** ultrasound frequency

However, higher frequency signals experience **stronger absorption** than the lower frequency signals

Image Resolution: B-Scan. 1 MHz

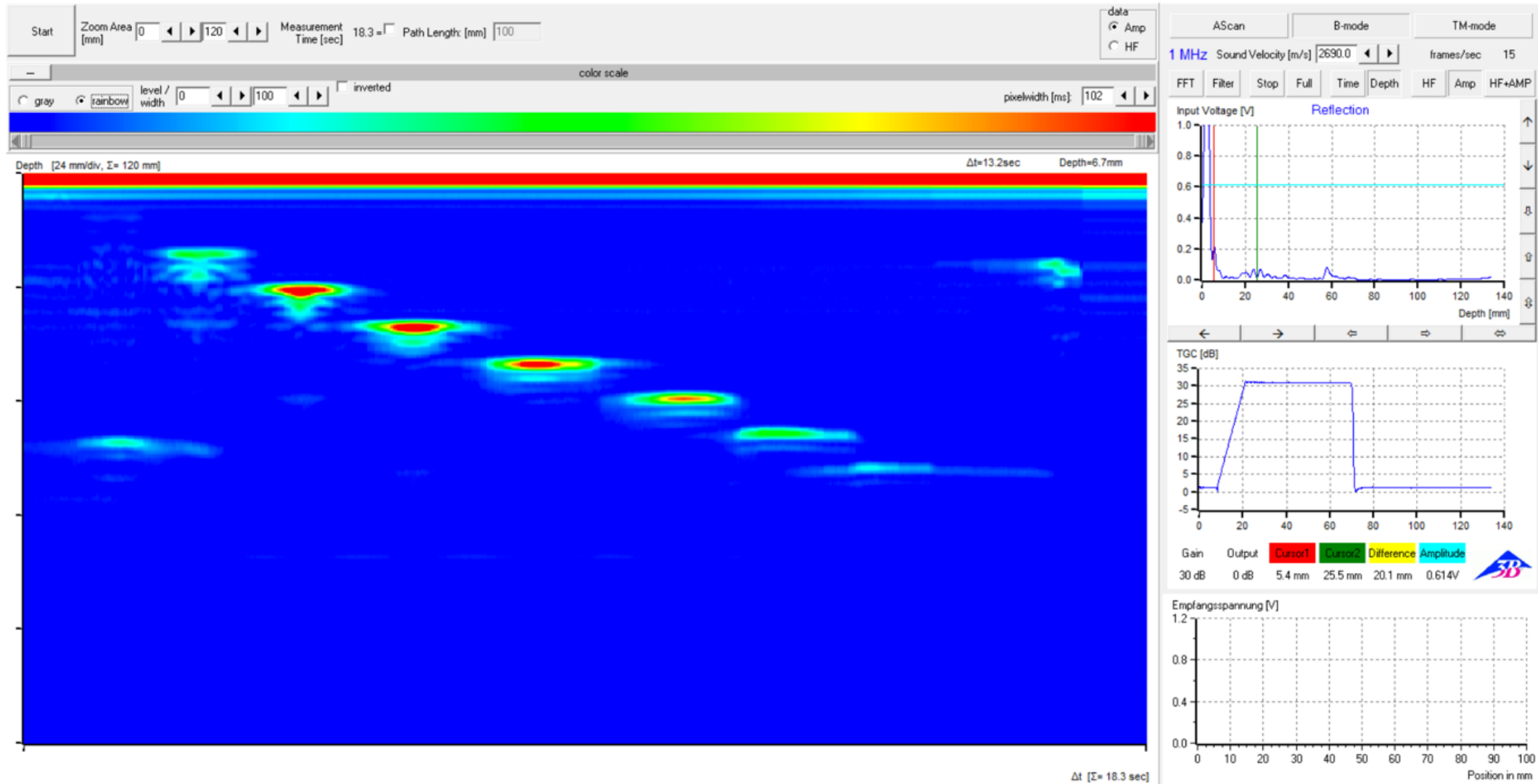
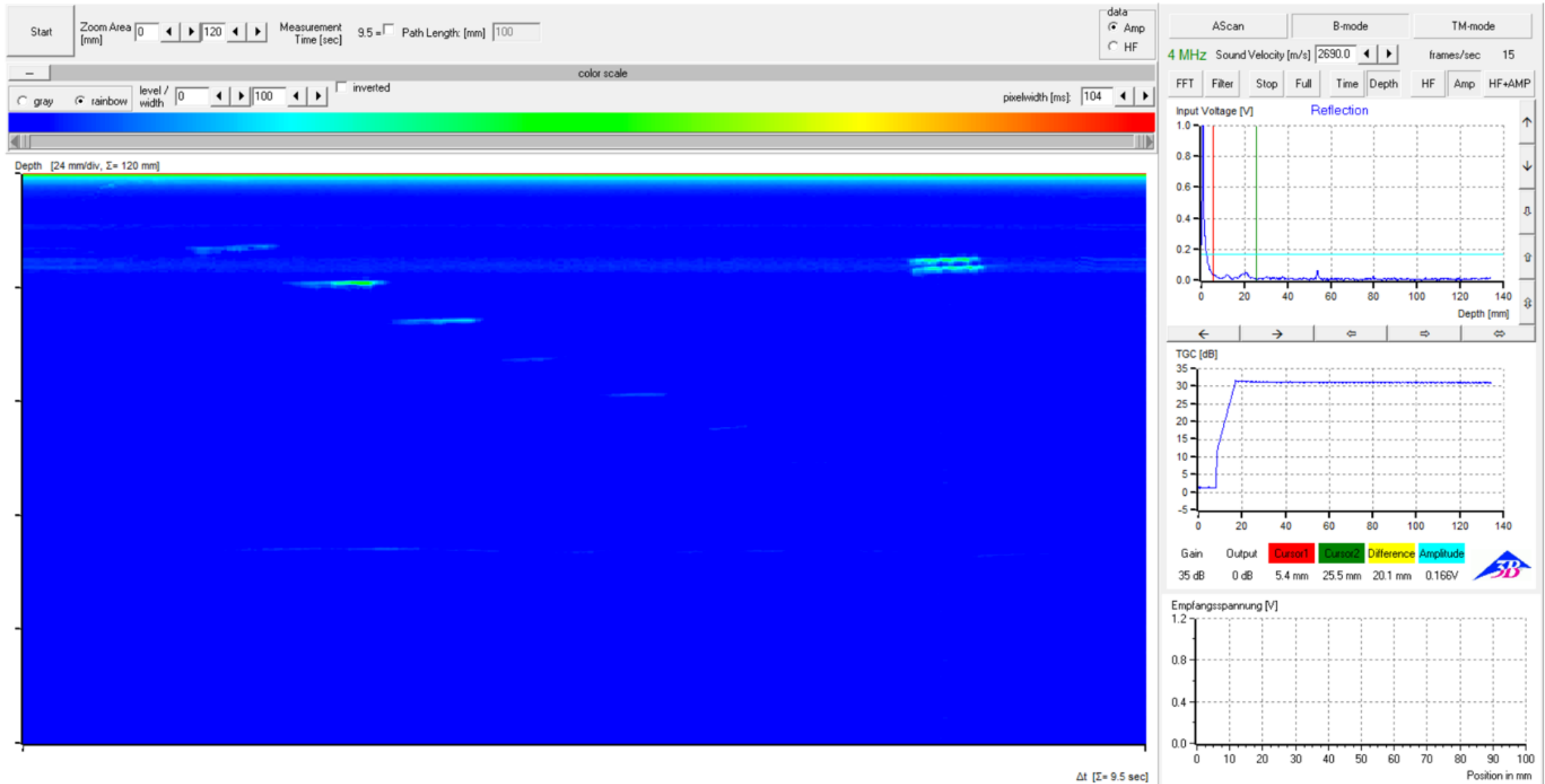


Image Resolution: B-Scan. 4 MHz





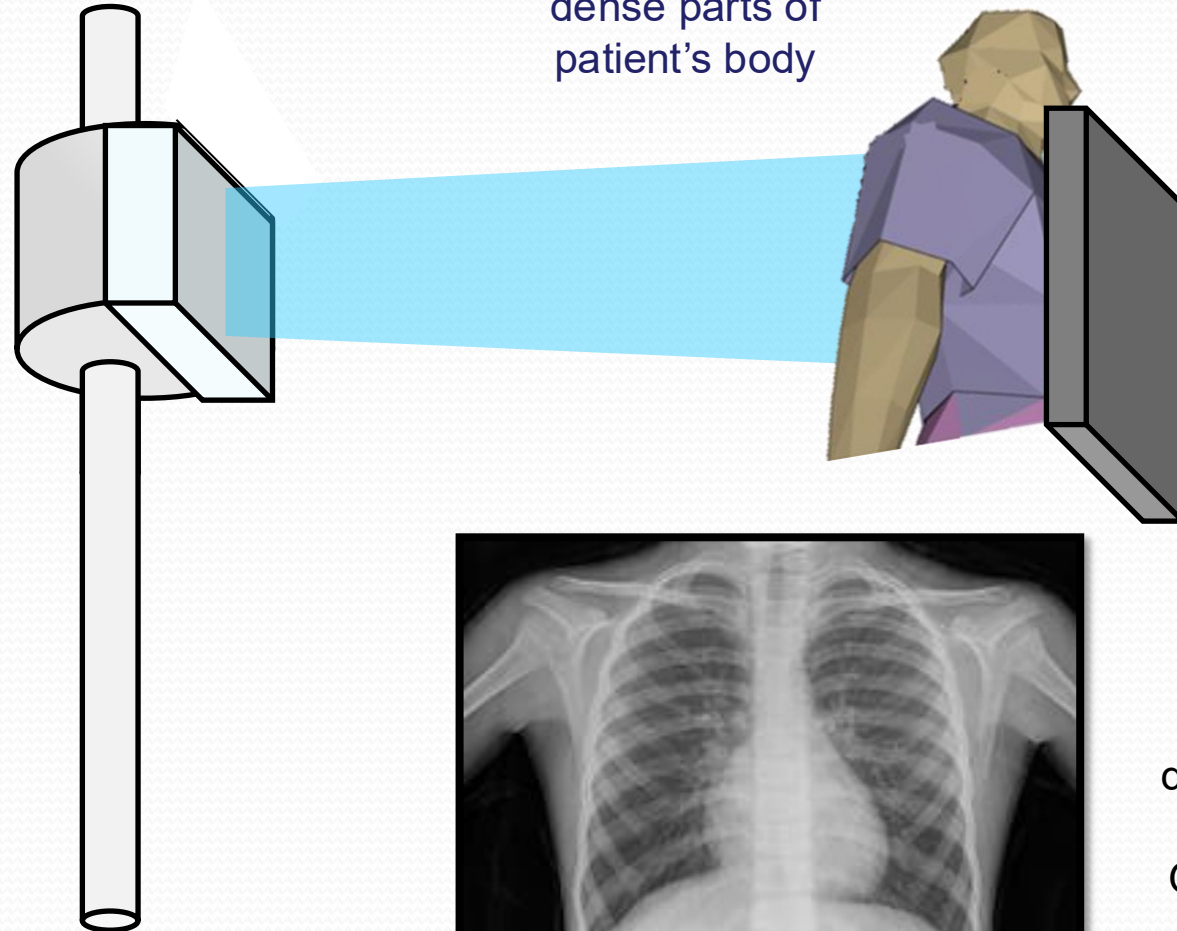
X-rays

Radiograph Formation

X-ray machine

X-rays absorbed by
dense parts of
patient's body

Photographic plate or
digital detector



White indicates
dense parts of body

Grey indicates less
dense regions

Nature of X-rays

X-ray radiation is a particular type of electromagnetic radiation

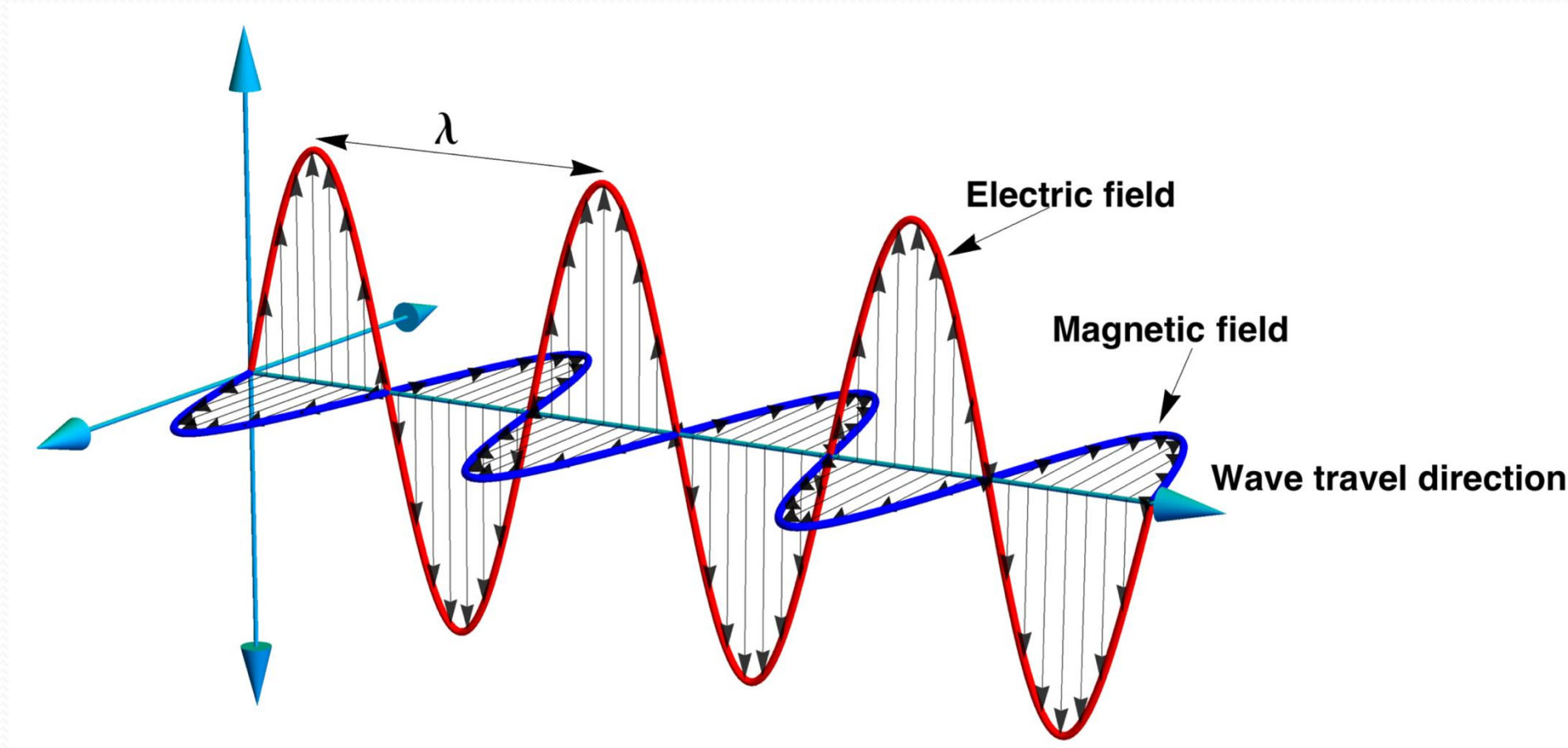
X-rays are characterized by:

Energy	E	(measured in electron-Volts)
Wavelength	λ	(measured in picometers, pm)
Intensity	I	(watts/m ²) (power per unit area)

Electromagnetic Wave

Electromagnetic waves consist of two components:

Oscillating **electric** and **magnetic** fields



Electromagnetic Waves

- An electromagnetic wave is a **transverse wave**: **electric** and **magnetic** components oscillate perpendicular to each other and perpendicular to the direction in which the wave travels
- Electromagnetic waves can travel through a vacuum as well as through matter (such as bones and tissues of the body)
- The speed of electromagnetic waves in a vacuum is equal to the speed of light in a vacuum

Speed of electromagnetic waves

$$c = f\lambda$$

c - speed (m/s)

f - frequency (1 Hz = 1/s)

λ – wavelength (m)

Speed of Light
in Vacuum

$$c \approx 300,000,000 \text{ m/s}$$

The Electromagnetic Spectrum

The classification of electromagnetic radiation with respect to **wavelength**

The seven parts of the electromagnetic spectrum:

radio waves, **microwaves**, infrared radiation,

visible light, ultraviolet radiation, **X-rays**, **γ-rays**

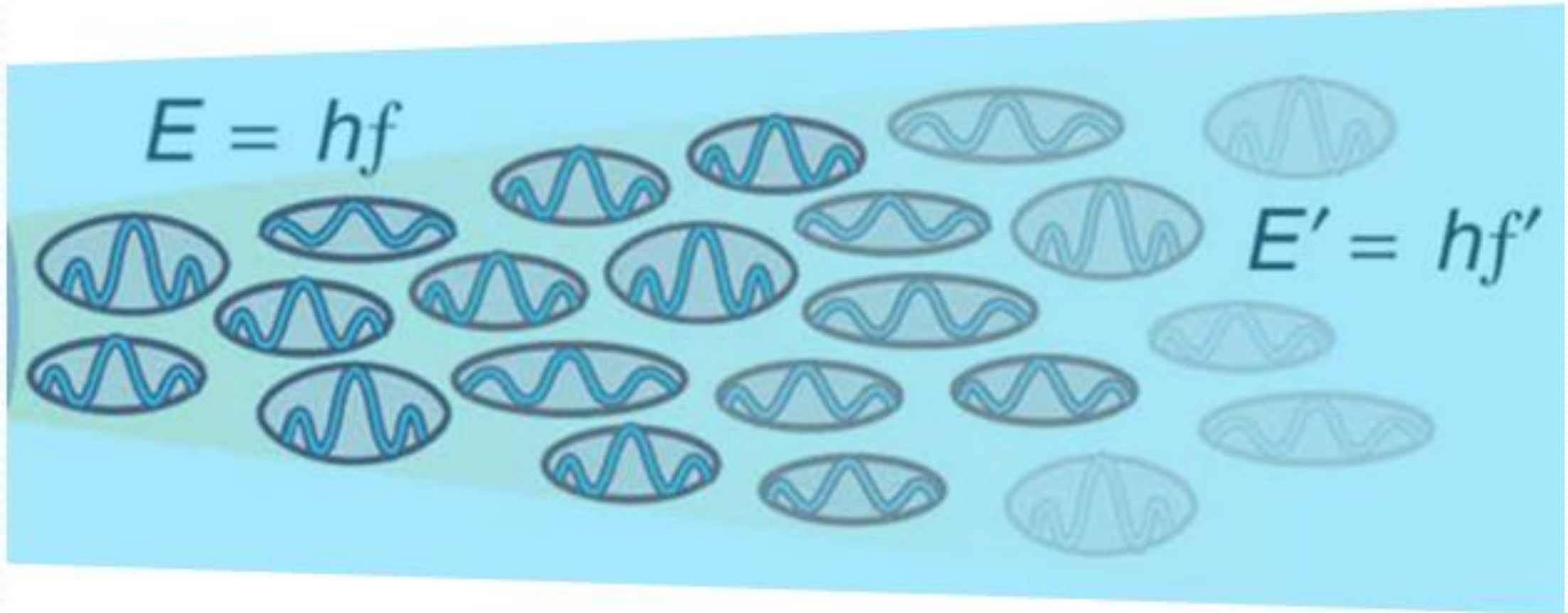
The Electromagnetic Spectrum

Type of Electromagnetic Radiation	Color	Wavelength
Radio waves		longer than 0.1 <i>cm</i>
Infrared waves		700 <i>nm</i> – 0.1 <i>cm</i>
Visible light	Red	630 – 700 <i>nm</i>
Visible light	Orange	590 – 630 <i>nm</i>
Visible light	Yellow	530 – 590 <i>nm</i>
Visible light	Green	480 – 530 <i>nm</i>
Visible light	Blue	430 – 480 <i>nm</i>
Visible light	Violet	400 – 430 <i>nm</i>
Ultraviolet		10 – 400 <i>nm</i>
X-rays and gamma rays		<i>less than 10 nm</i>

X-rays:

**Invisible, penetrating form of
very high frequency (short wavelength)
electromagnetic radiation,
following the ultraviolet range
in the electromagnetic spectrum**

X-ray Beam – Streams of Photons



Planck-Einstein Relation

Relation between the energy and the frequency
of an X-ray photon

$$E = hf = \frac{hc}{\lambda}$$

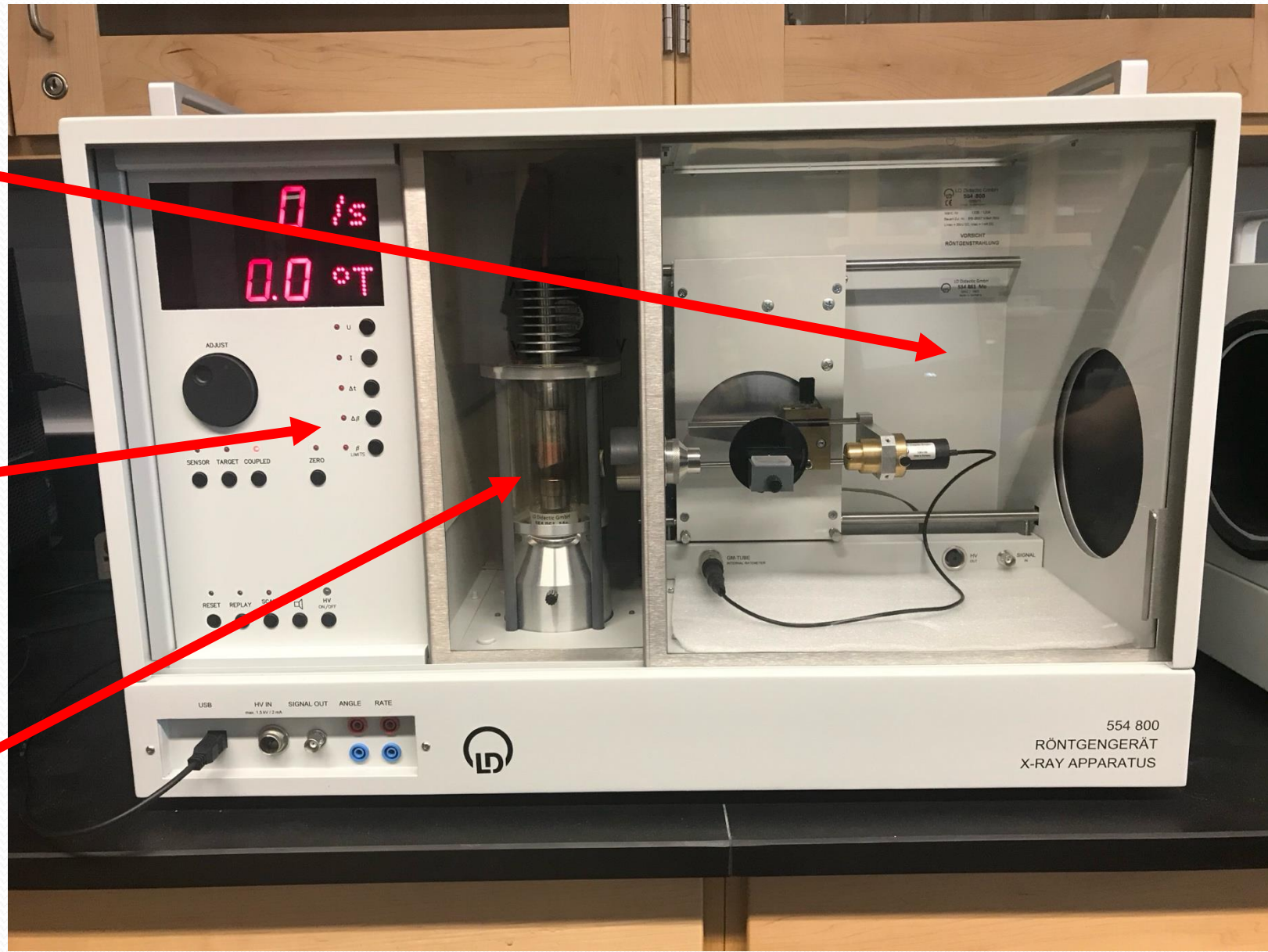
Planck's constant: $h = 4.136 \times 10^{-15}$ (eV ·
sec)

Laboratory X-ray Apparatus

Experimental Chamber

Control Panel

X-ray Tube



X-ray Tube



X-ray Tube

A device for generating X-rays

Main components:

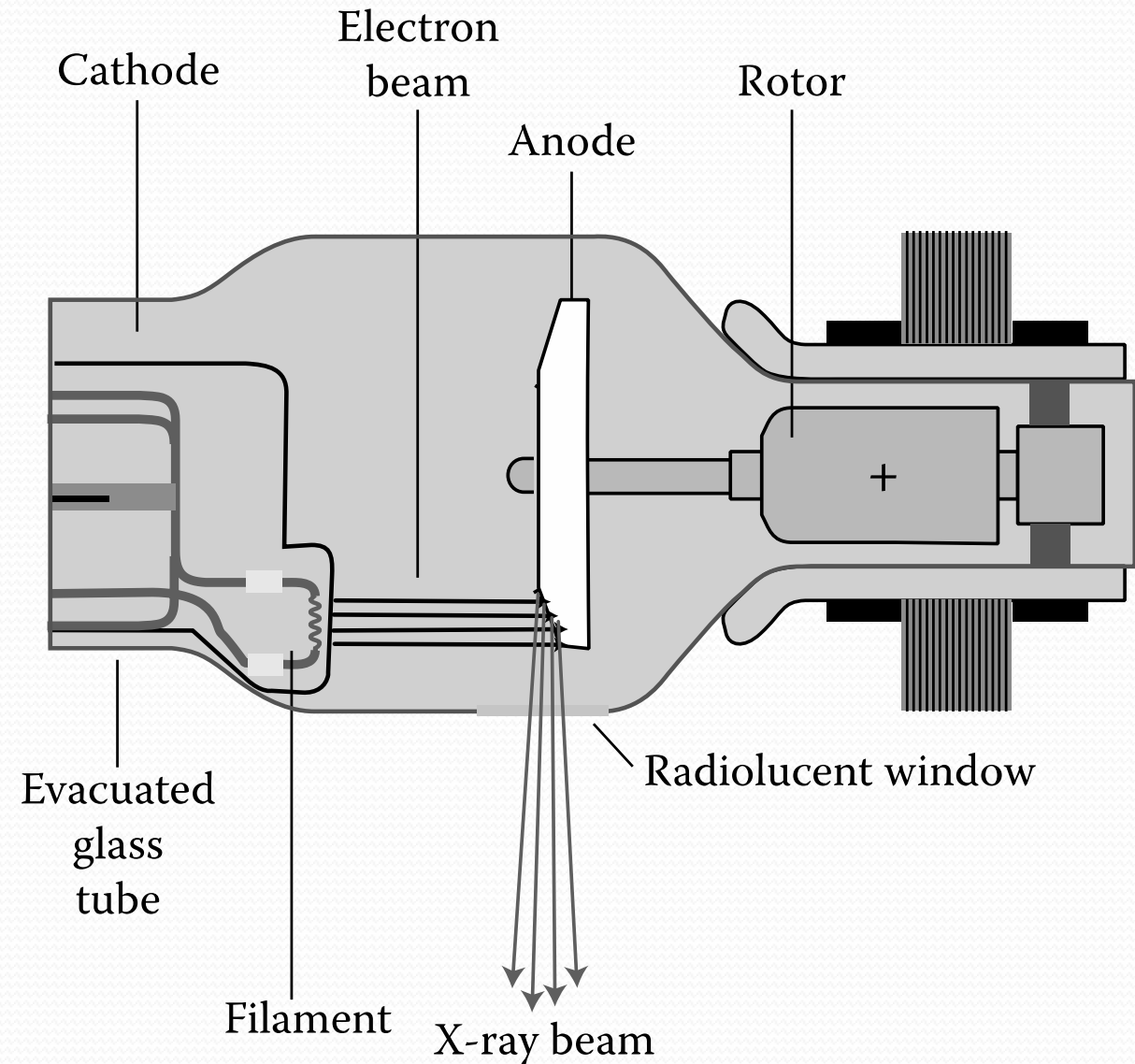
Two electrodes: cathode and anode

Heavy metal target in the anode

Metal filament in the cathode

Cooling system for the anode

The components are housed in an evacuated glass tube

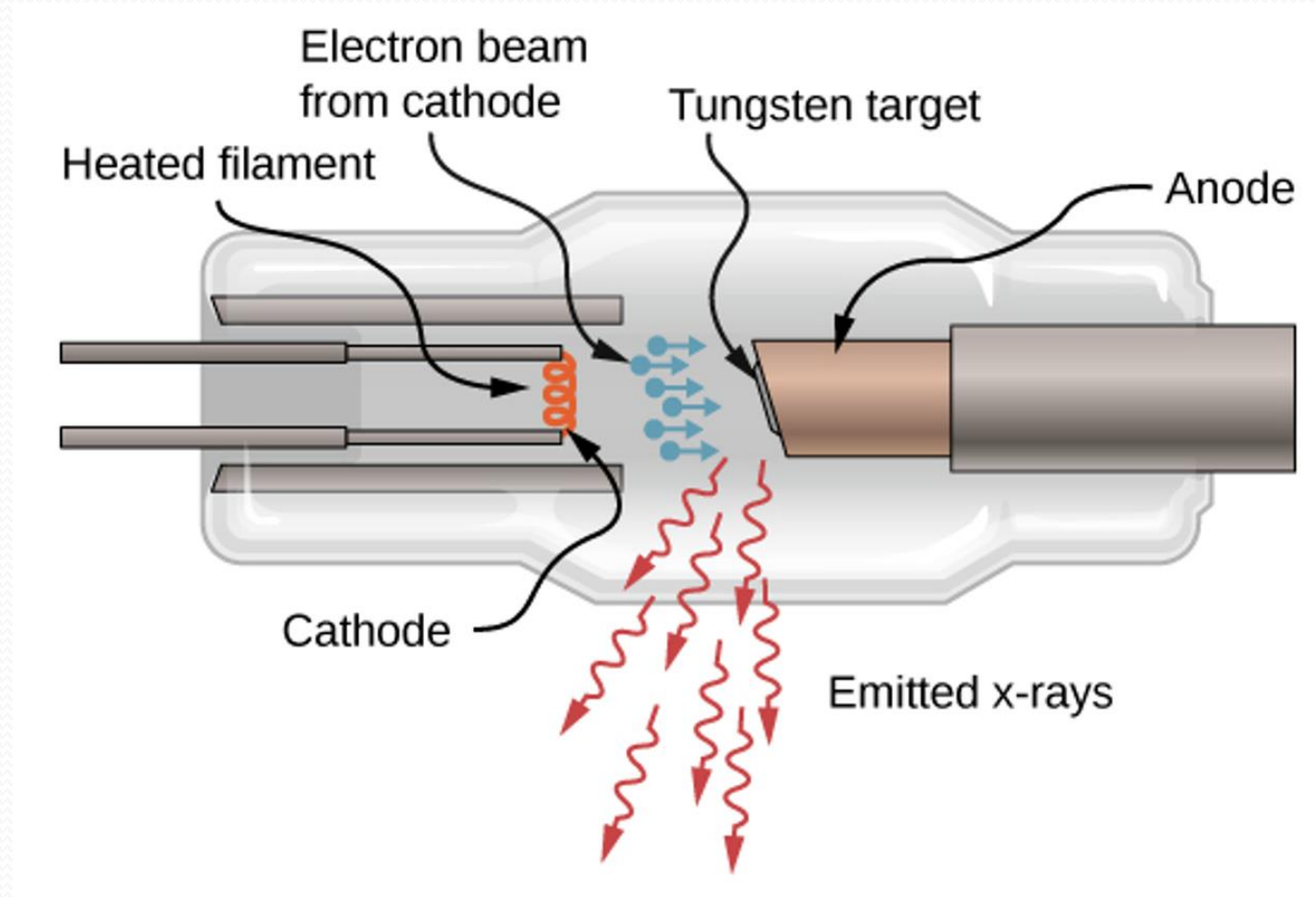


The emission current in the **filament** heats the **filament** to very high temperatures

Thermionic emission - electrons in the hot filament are **emitted** into the space between the cathode and the anode

Emitted electrons are accelerated by the **high voltage** between the **cathode** and **anode**

X-ray Tube

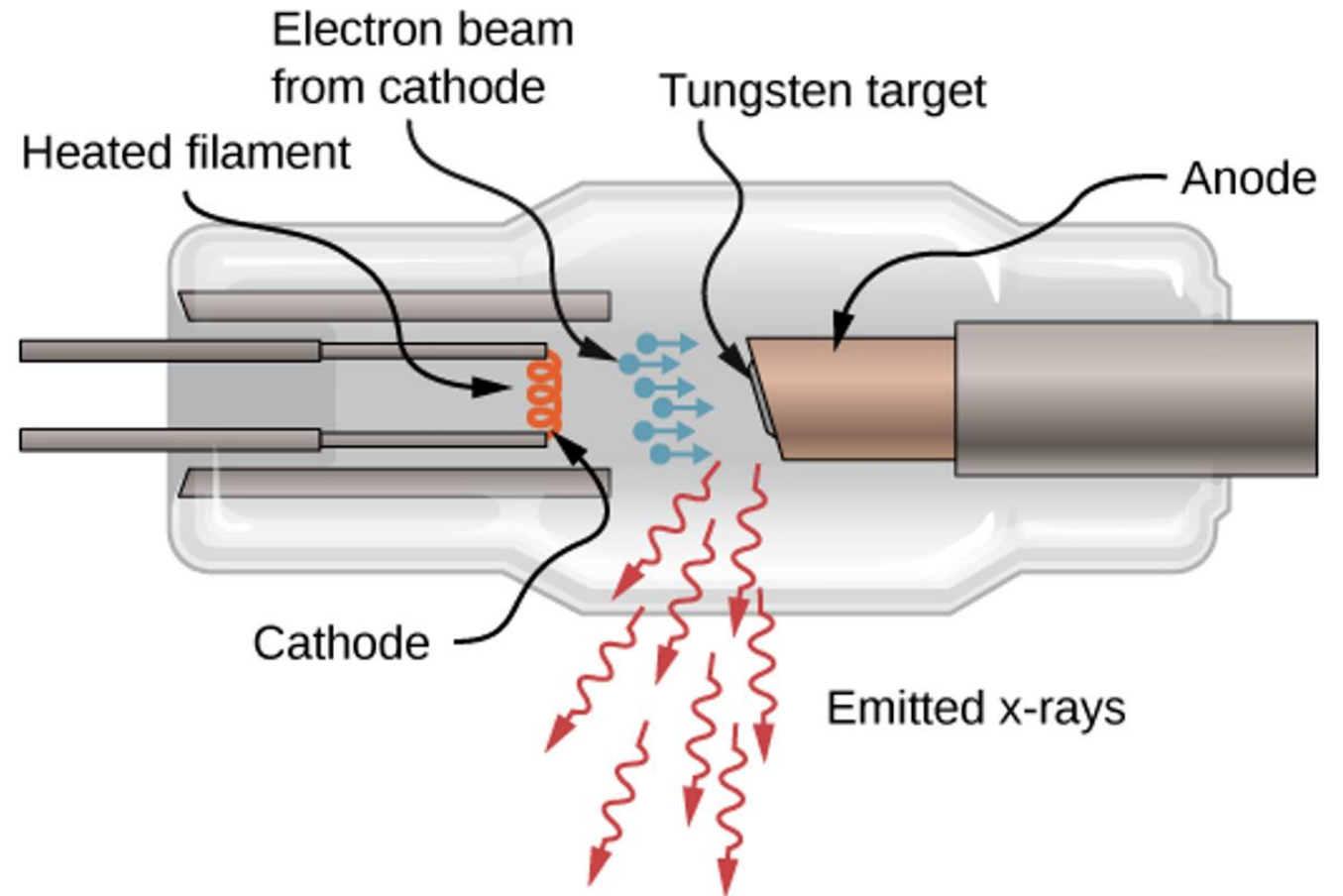


X-ray Tube: X-ray emission

High energy electron beam strikes the metal target in the anode

Inside the metal target the electrons are slowed down by the atoms of the anode

The energy of electrons is converted into X-ray radiation and heat



Bremsstrahlung Radiation

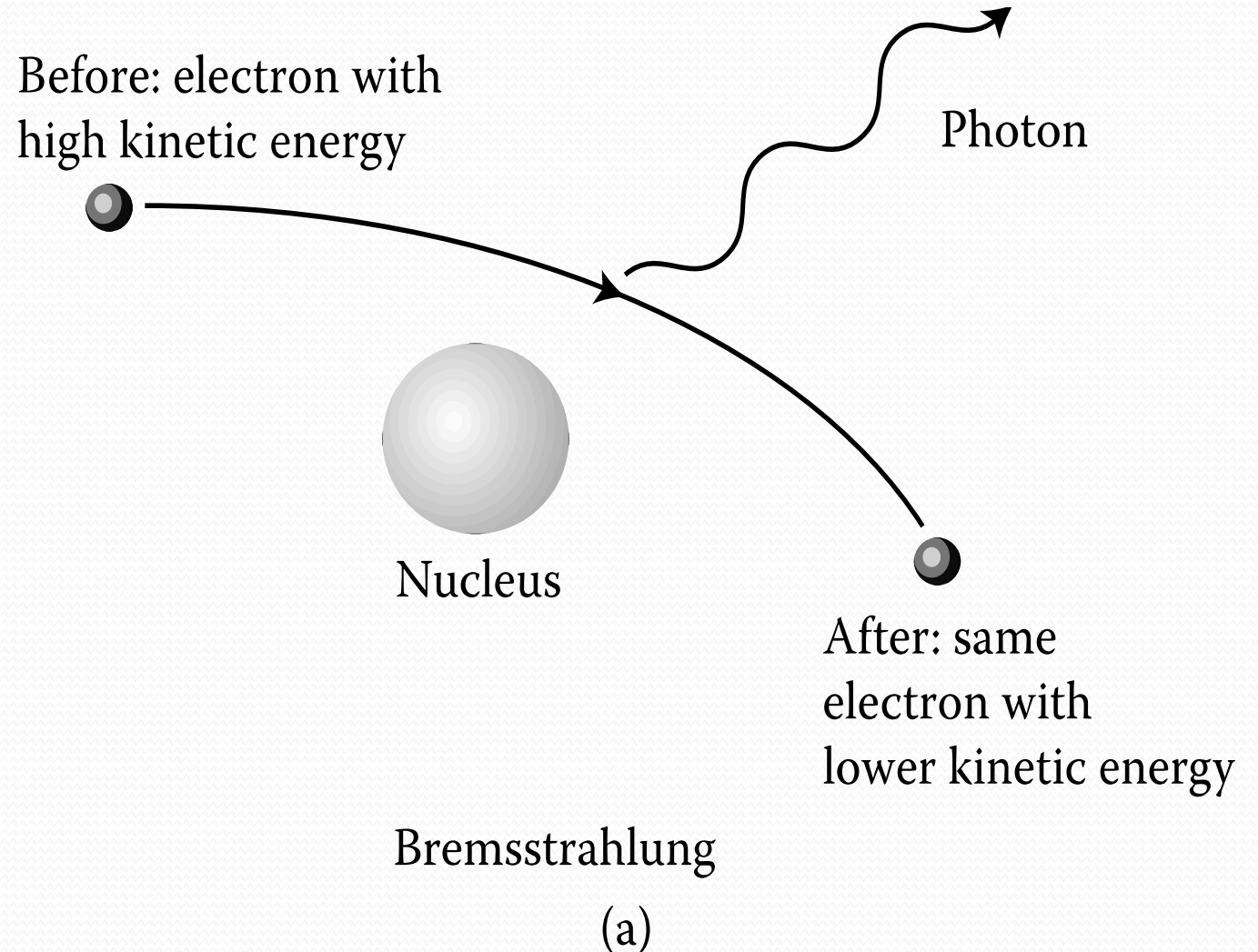
Bremsstrahlung (braking radiation) is one of the processes in which X-ray photons are created when the metal target of the anode is bombarded by high energy electrons.

An electron passing by one of the atoms of the anode is **slowed down** and deflected by an attractive force from an oppositely charged nucleus.

The **energy lost** by an electron is carried away by an X-ray photon.

Bremsstrahlung Radiation

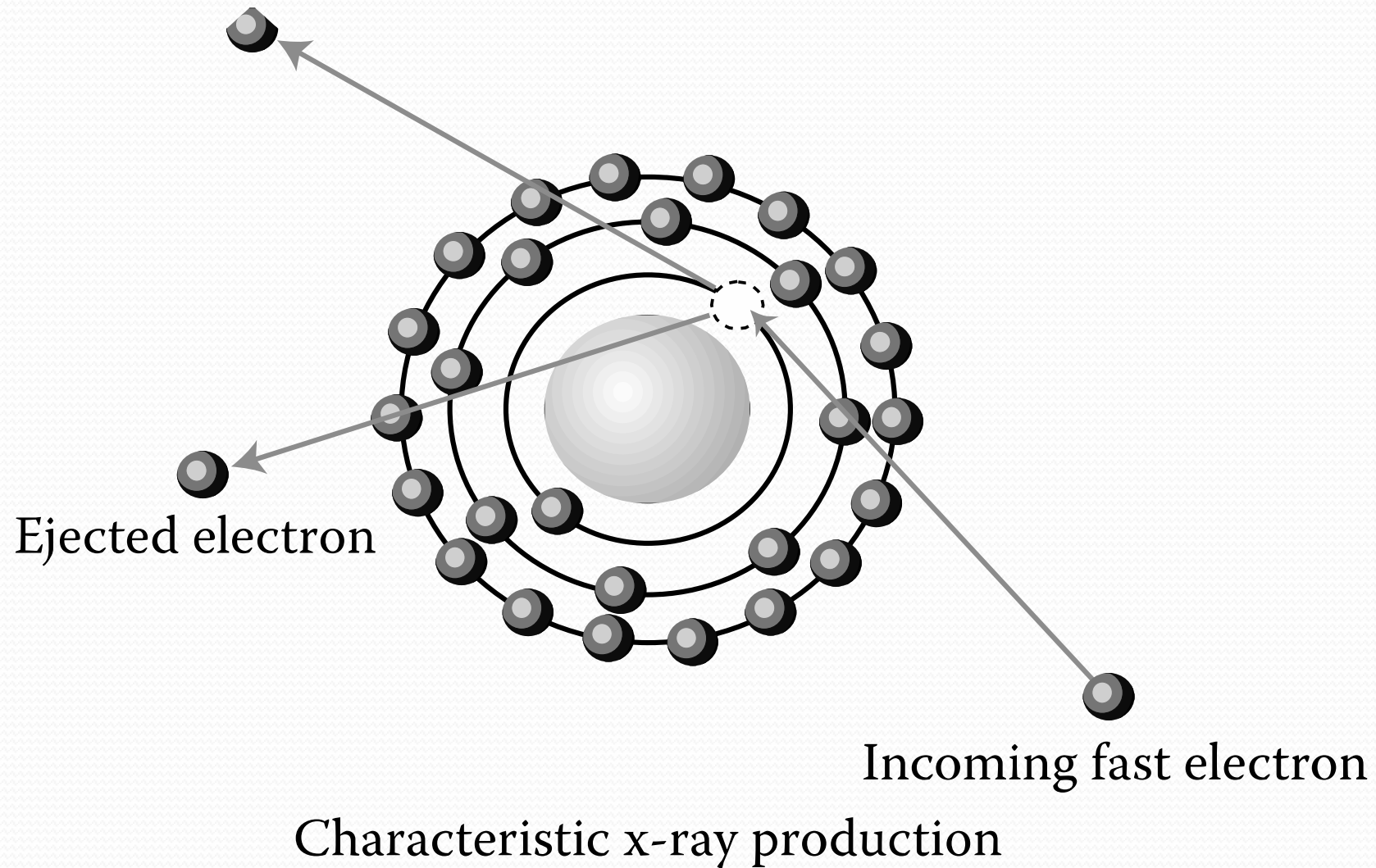
Energy losses by electrons as it slows down are not restricted to any discrete value; as a result, the bremsstrahlung X-rays have a **continuous range of wavelengths**



Characteristic X-ray Radiation

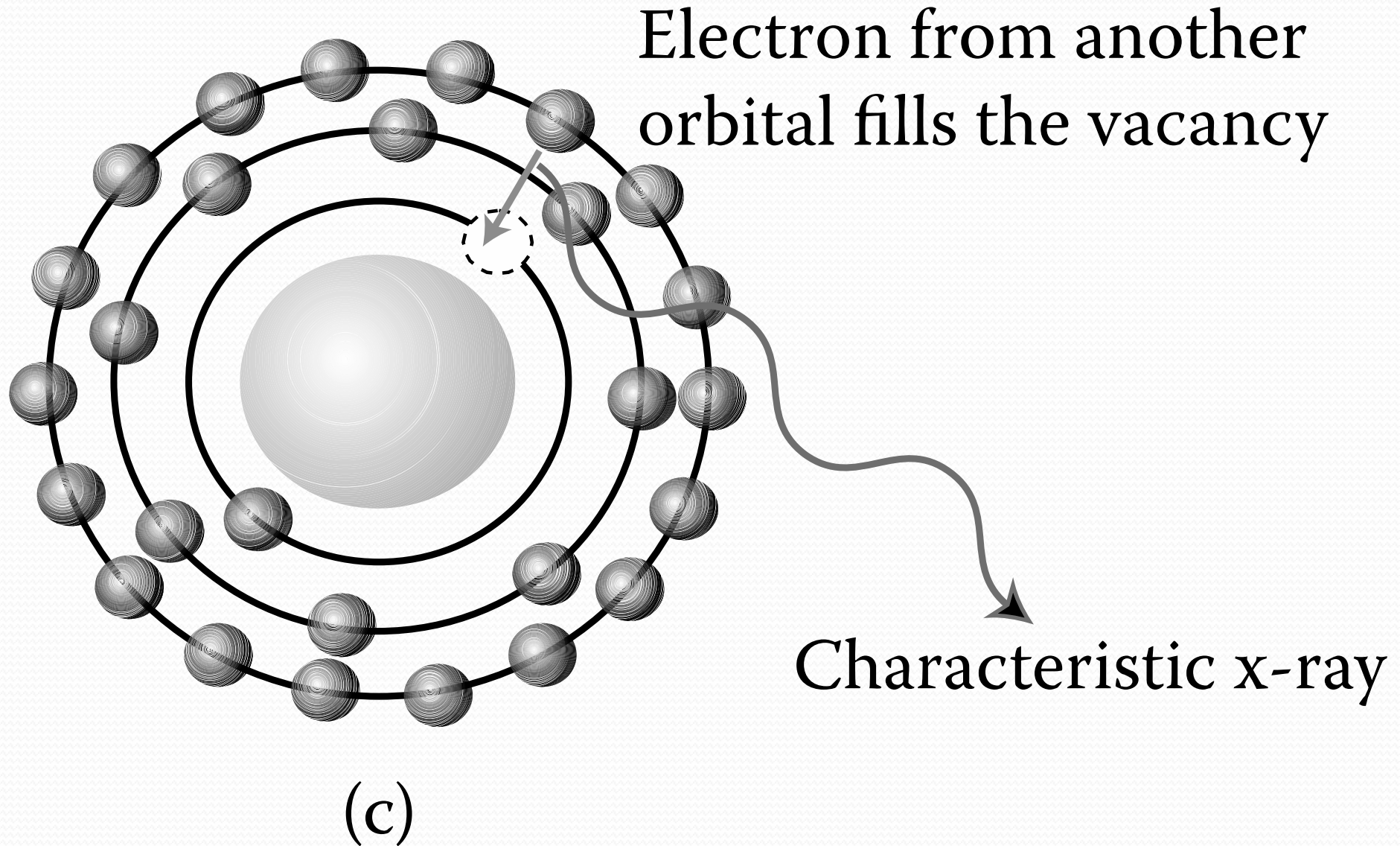
- The shell structure of atoms allows for another type of X-ray emission called characteristic X-ray radiation
- In this process, the **incoming electron**, instead of emitting an X-ray photon, collides with an electron in one of the **inner energy shells**.
- If the **incoming electron** has enough energy, this collision may result in the ejection of the atomic electron and creation of a **vacancy** in that shell.

Characteristic X-ray Radiation



(b)

Characteristic X-ray Radiation



Characteristic X-ray Radiation

- Because an inner shell corresponds to lower energy, one of the electrons from a higher energy shell makes a transition to the lower energy shell from which is missing one electron
- As a result of this transition, an amount of energy equal to the energy difference, E , of the initial and final shells is released in the form of an X-ray photon

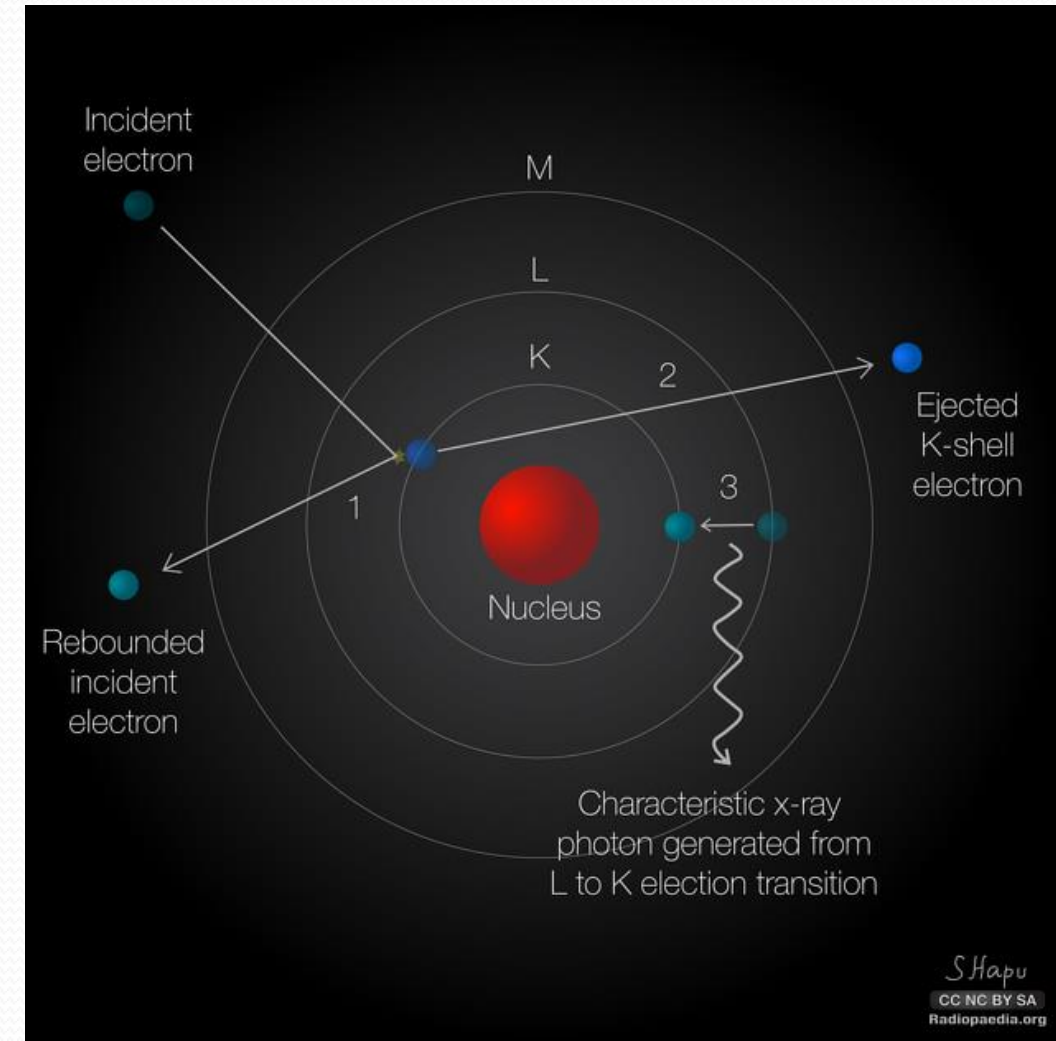
Characteristic Radiation

Using Planck-Einstein relation, the wavelength of the emitted X-ray is

$$\lambda = \frac{hc}{E}$$

E is the energy difference

Each chemical element has its own discrete set of wavelengths that can be emitted in this way – this radiation is called characteristic radiation





X-ray Tube Emission Spectrum

Tube Ratings

kVp – peak-kilovoltage	U (kV)
mA – milliamperage	I (mA)
Exposure (seconds)	Δt (<i>sec</i>)

X-ray Tube Emission Spectrum

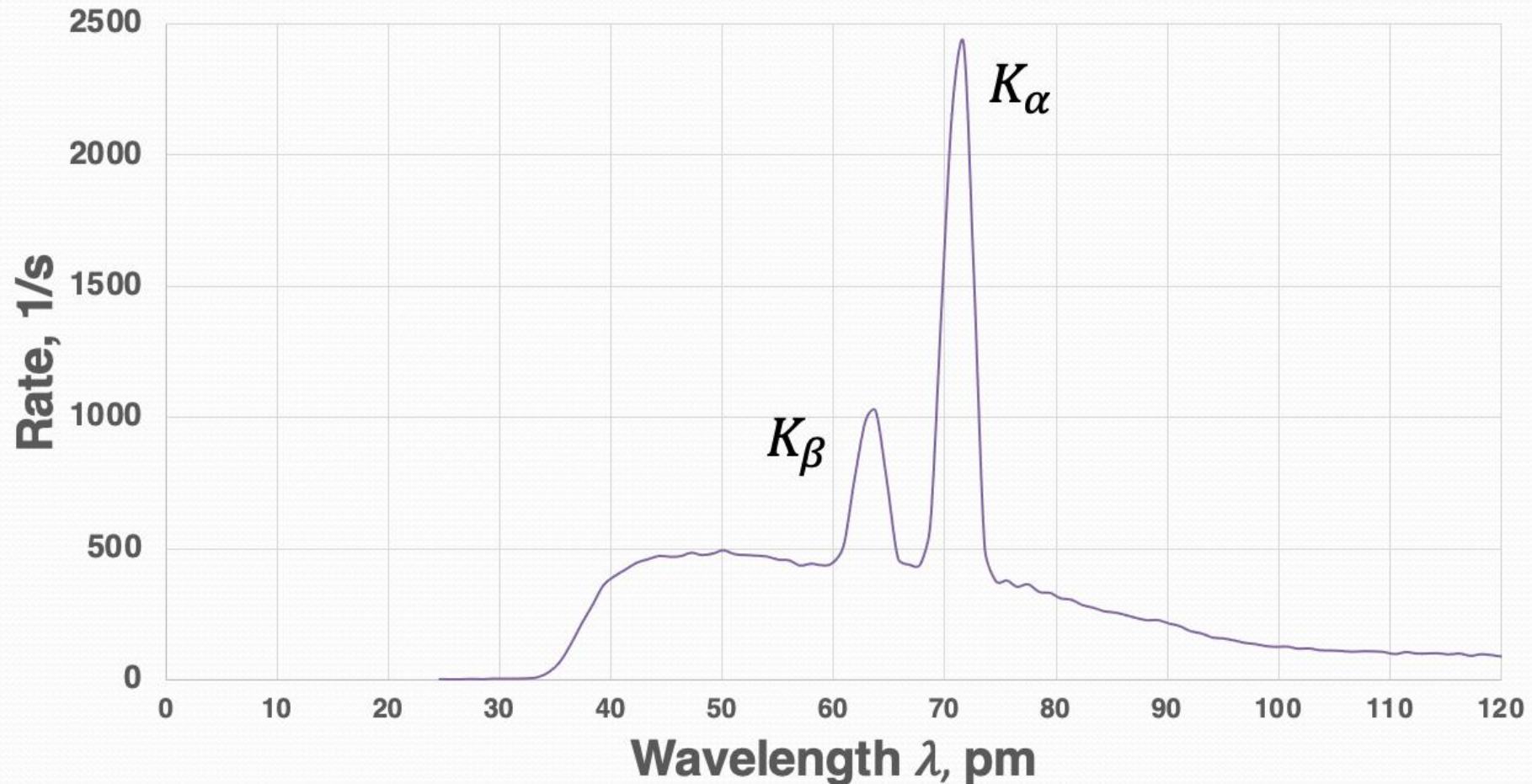
An X-ray tube is characterized by its **emission spectrum** (also called the **energy spectrum**).

An X-ray tube **emission spectrum** can be visualized as a plot of the **intensity** (or the emission rate) versus the **wavelength** (or **energy**) of the emitted X-rays.

X-ray Tube Emission Spectrum

Plot of the Emission Rate versus X-ray wavelength

Molybdenum Anode $U = 35 \text{ kV}$ and $I = 1 \text{ mA}$



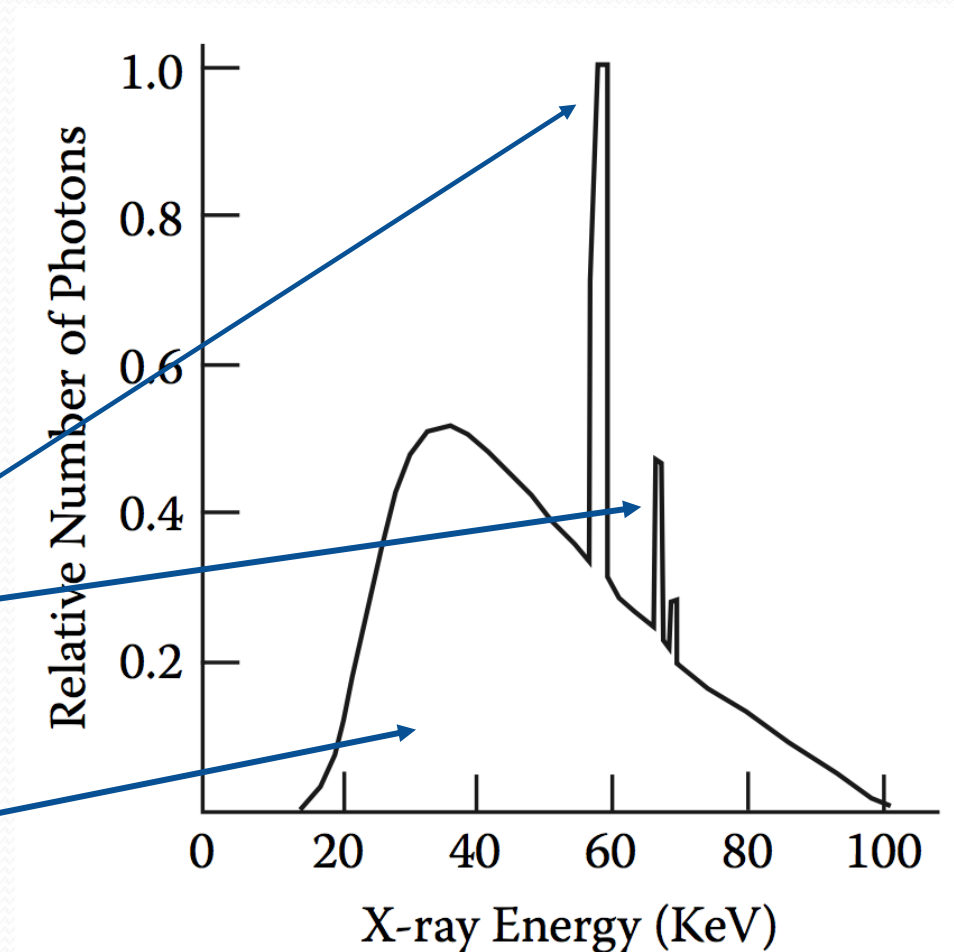
X-ray Tube Emission Spectrum

Emission spectrum of an X-ray tube with a **tungsten anode** for 100 kV voltage with a **2.5 mm aluminum filter**

The emission spectrum has two components: **continuous bremsstrahlung** and **discrete characteristic X-ray radiation**

Characteristic radiation

Bremsstrahlung radiation





Interaction of X-rays with Matter

X-ray Absorption in Tissues

- The intensity (or the X-ray rate) of an X-ray beam depends on the total number of the X-ray photons in the beam
- When an X-ray beam passes through tissues the X-ray photons interact with atoms of the tissues
- For the formation of an X-ray image two types of processes are important: photoelectric absorption and Compton scattering

Interaction of X-rays with Atoms

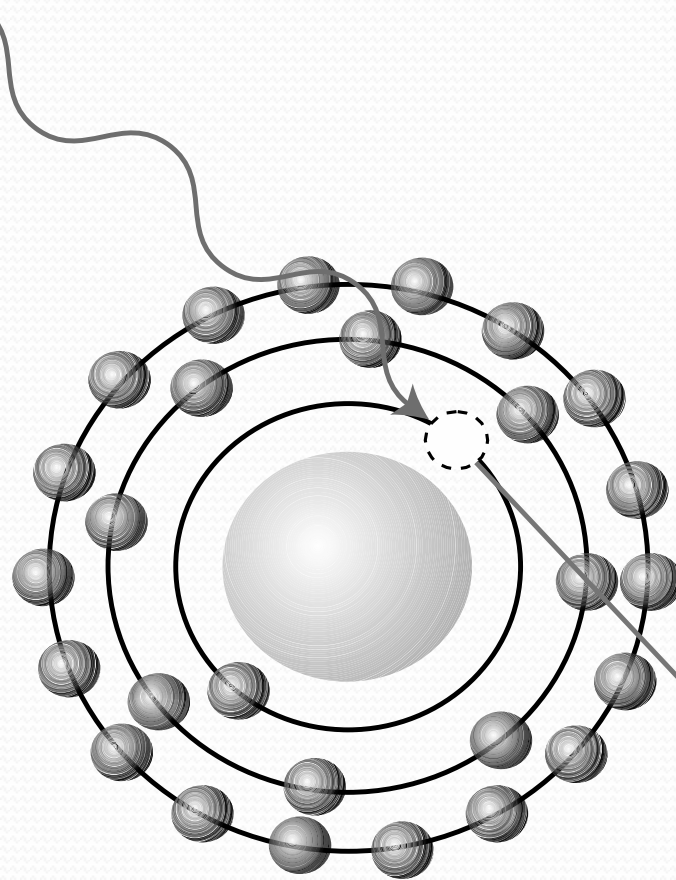
- **Photoelectric effect**
- **Compton Scattering**

Photoelectric Effect

- An incoming X-ray photon is absorbed by an atom
- One electron is ejected from the atom – the atom is ionized
- The photoelectric effect is dominant for X-rays with energies **below 25 keV**
- The photoelectric effect depends on the tissue type
 - Atoms in soft tissues: **C, O, H, Na**
 - Atoms in bones: **Ca**

Photoelectric Absorption

Incoming
x-ray photon

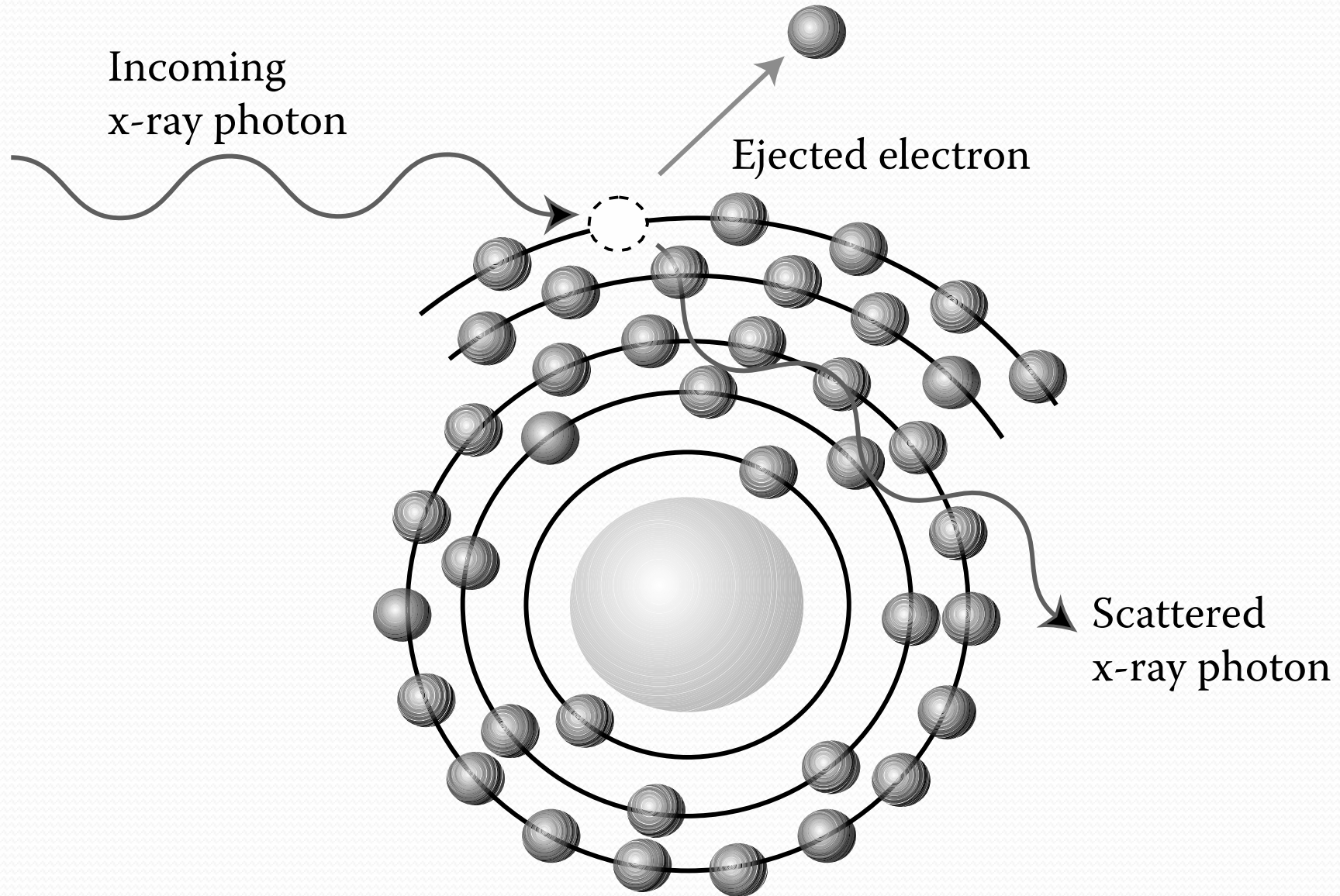


Photoelectron

Compton Scattering

- An incoming X-ray photon is **scattered** by an atom –
The energy of scattered photon is less than the energy of the incoming photon
- Compton scattering is dominant for X-rays with energies **above 25 keV**
- Compton scattering is relatively independent of the tissue type
- Compton scattering reduces contrast in X-ray imaging

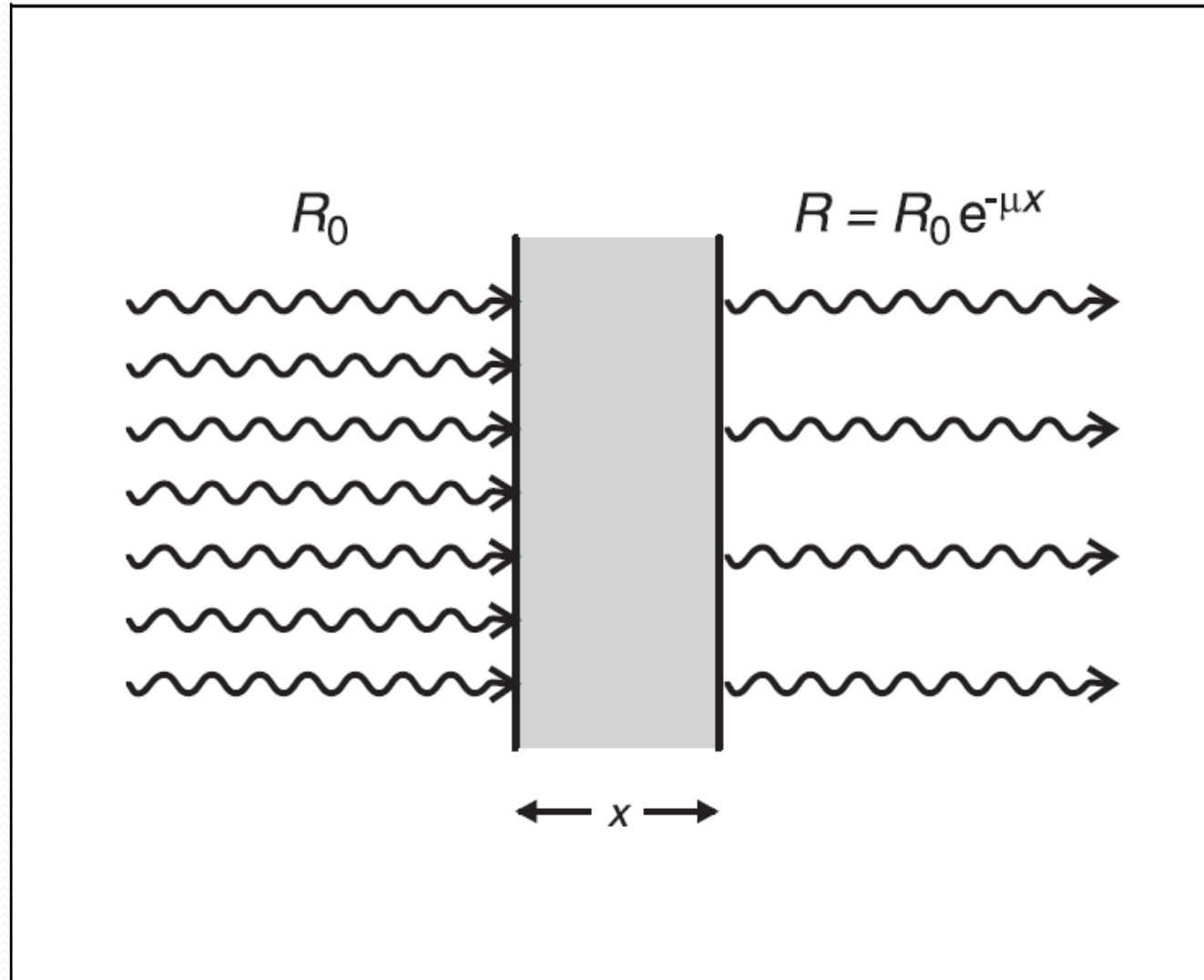
Compton Scattering





X-ray Attenuation Coefficient

X-ray Absorption in Tissues



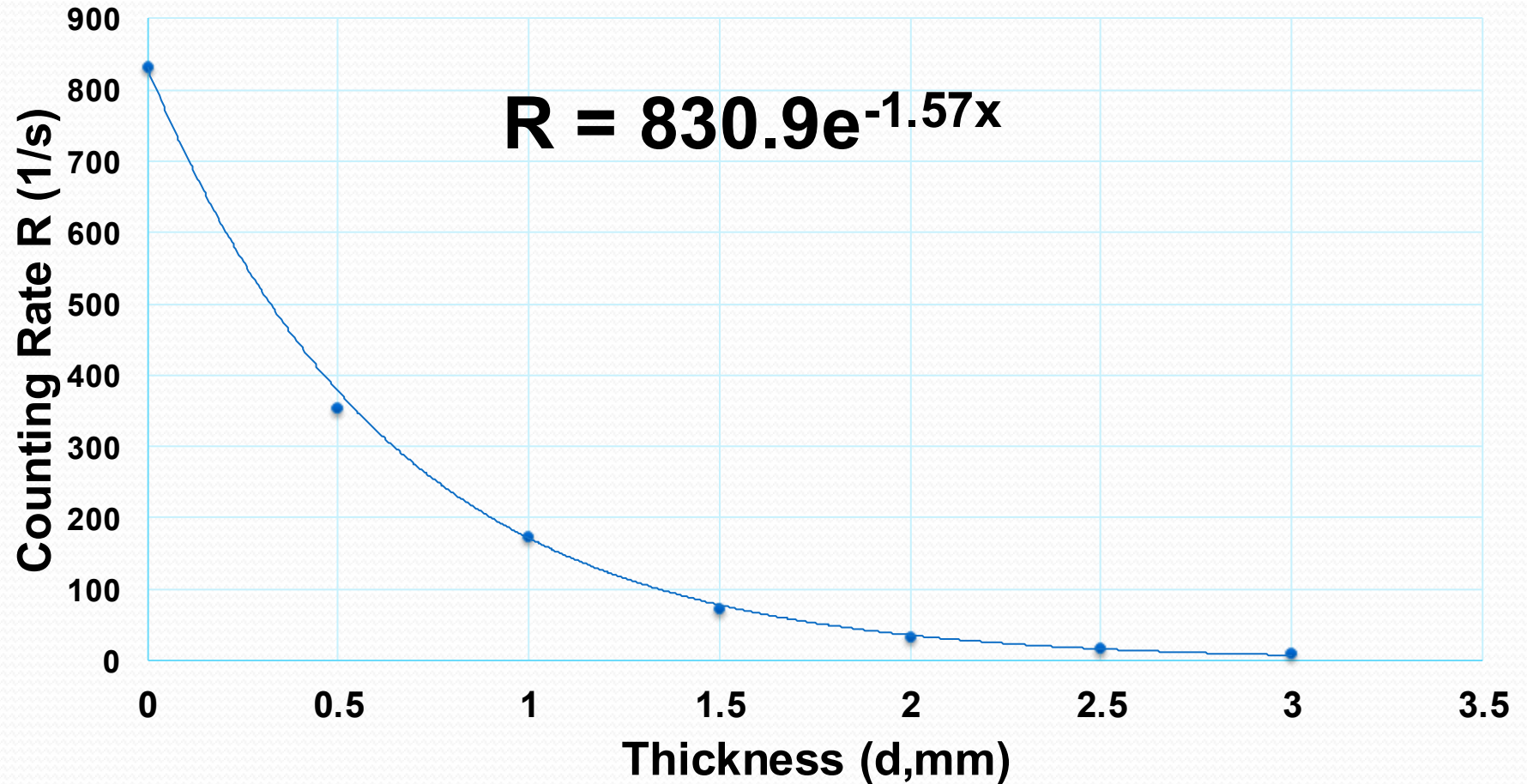
Exponential Decrease of Intensity

$$R = R_0 e^{-\mu x}$$

Attenuation
Coefficient:

$$\mu = 1.57 \text{ mm}^{-1}$$

Attenuation Rate with Zr Filter



X-ray Attenuation: Transmittance

$$I = I_0 e^{-\mu x}$$

$$T = \frac{I}{I_0} = e^{-\mu x}$$

I_0	Initial intensity
I	Final intensity
T	Transmittance
x or d	Tissue thickness
μ	X-ray attenuation coefficient

X-ray Attenuation Coefficient

Absorbing tissues and materials	μ, cm⁻¹ 60 keV X-rays
Fat	0.1788
Muscle, body fluids	0.2045
Water	0.2055
Brain	0.2061
Bone	0.466 – 0.548
Air	~0.0003

Absorption Contrast

Depends on

- The energy of the x-ray beam
- The tissue thickness
- The tissue mass attenuation coefficient
- Degree of X-ray scattering (noise)

Mass Attenuation Coefficient

$$\frac{\mu}{\rho} \sim Z^3$$

Increases as cube of the atomic number

μ – coefficient of attenuation for a given tissue

ρ – the density of the tissue

Computed Tomography

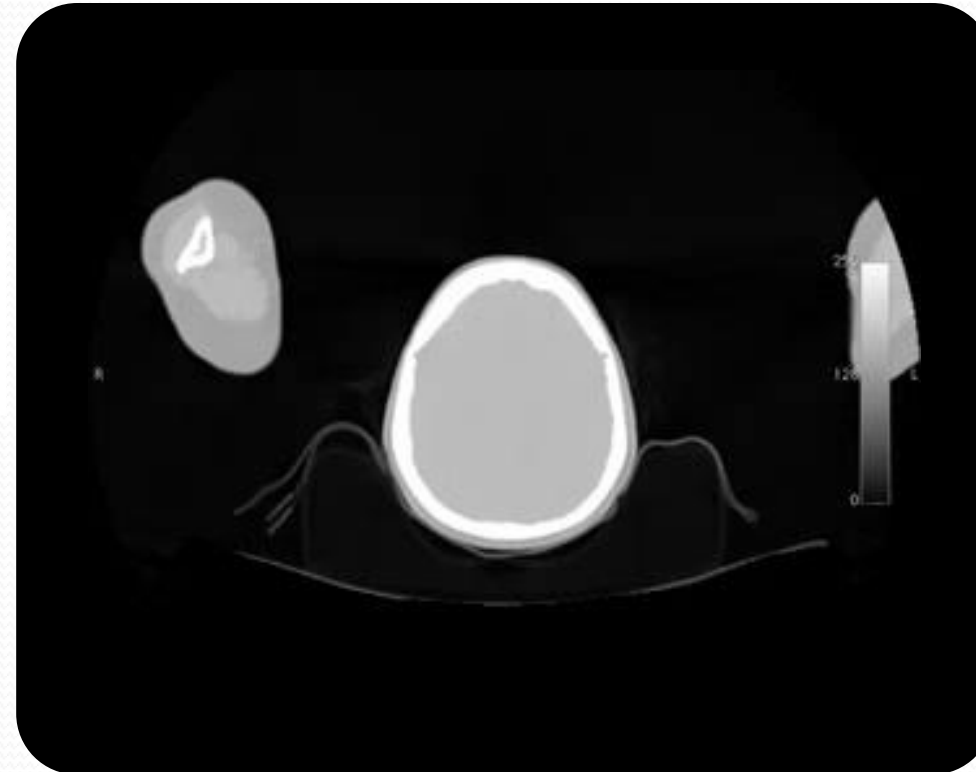
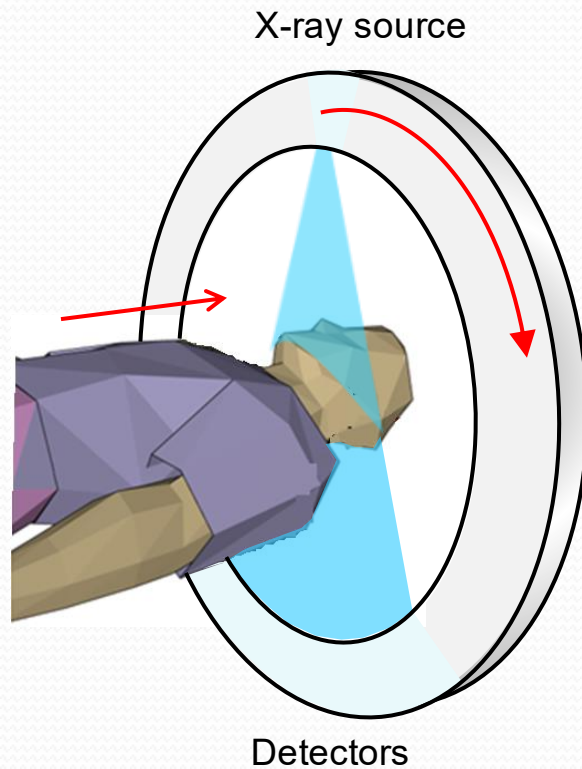
A procedure in which many **X-ray projections** are combined, and **two-dimensional images** are reconstructed using a computer algorithm

Computed Tomography

- **Axial Projections**
- **Filtered Back Projection**
- **Gray Scale based on Hounsfield Units**



CT: Axial Projections



Play



MRI

Magnetic Resonance Imaging

Magnetic resonance imaging (MRI) is a non-invasive imaging technique that uses **magnetic fields** and **radiofrequency waves** to induce protons to **emit radio-wave signals** that are detected and used to generate two and three-dimensional images of a living subject in the MRI scanner.

MRI Machine

A large magnetic field is generated by the **cylindrical solenoid** surrounding a patient

(credit: Liz West)





MRI Machine

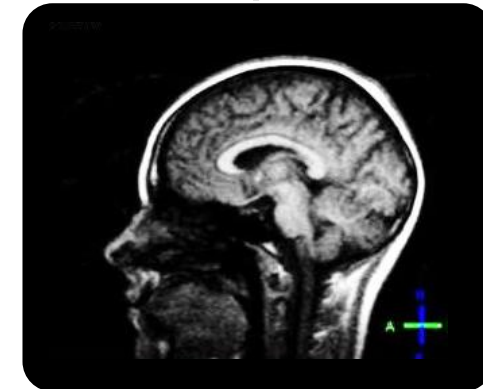
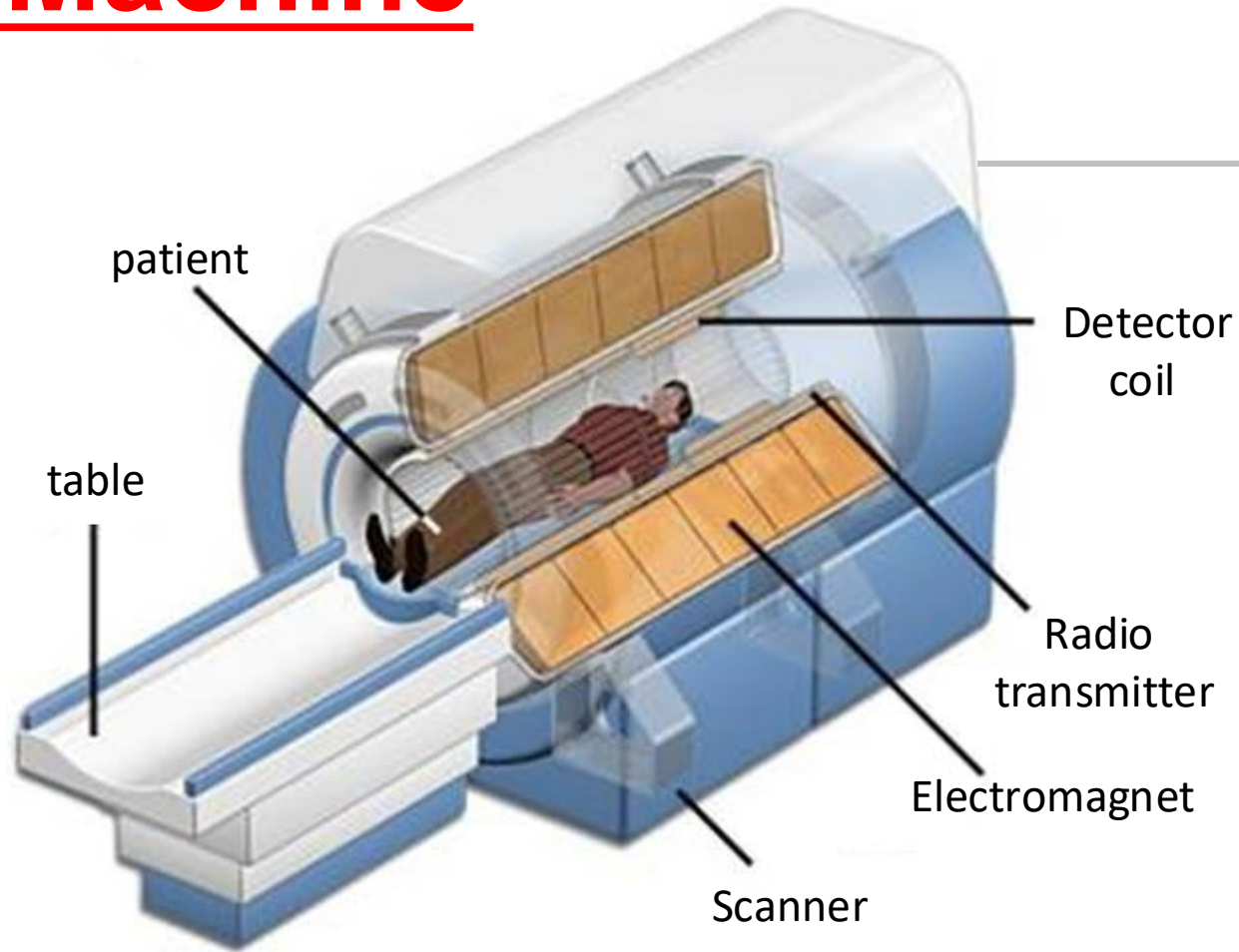


Image courtesy of Pancreatic Cancer Action

Magnetic Field Strength (Magnitude)

A magnetic field produced by a source is characterized by its **direction** and its **strength** (or **magnitude**) B_0

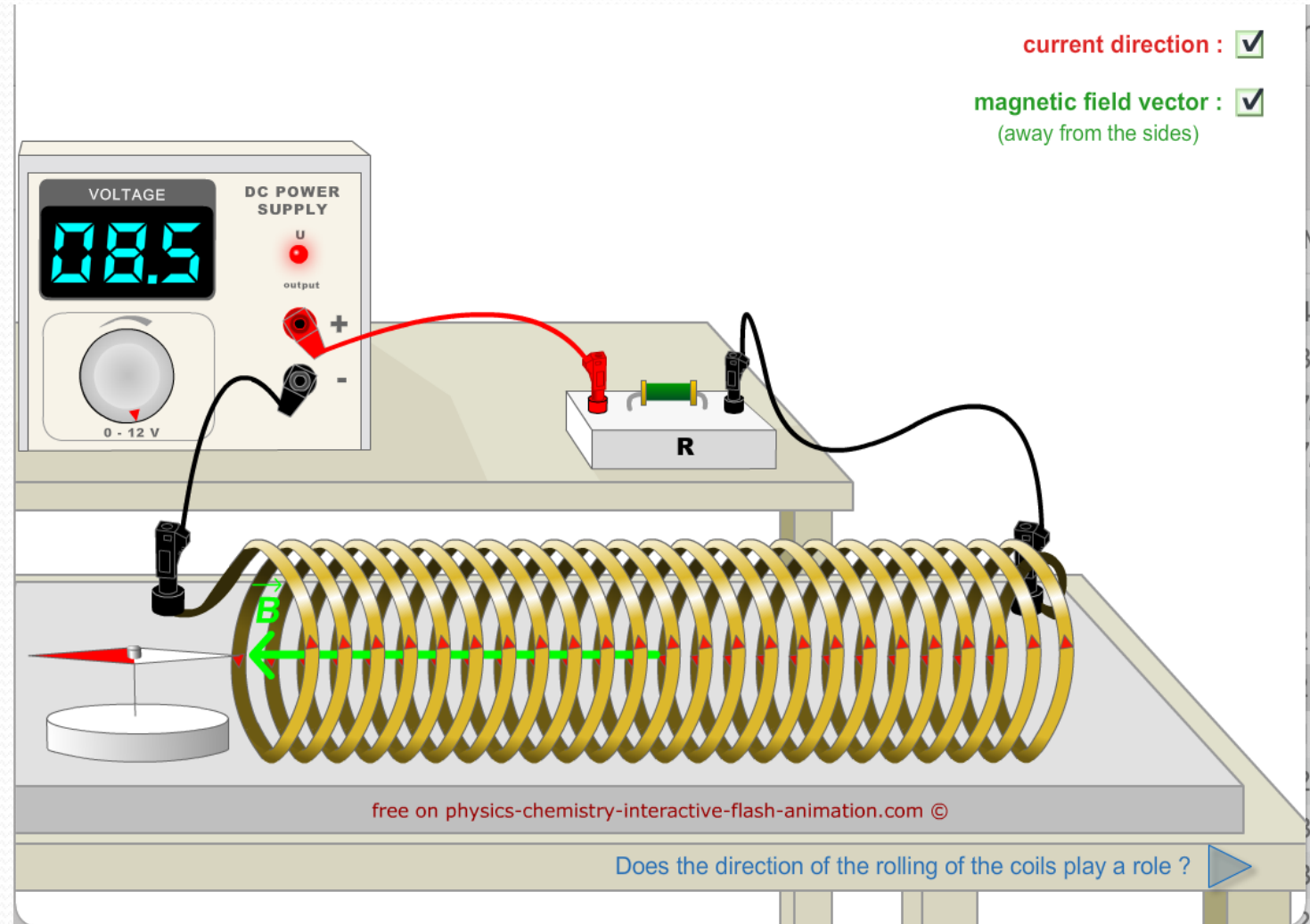
The **direction** at each point around the source is given by the magnetic field lines

The **strength** of a magnetic field is measured using the unit **Gauss** or **Tesla**

$$1 \text{ Tesla} = 10000 \text{ Gauss}$$

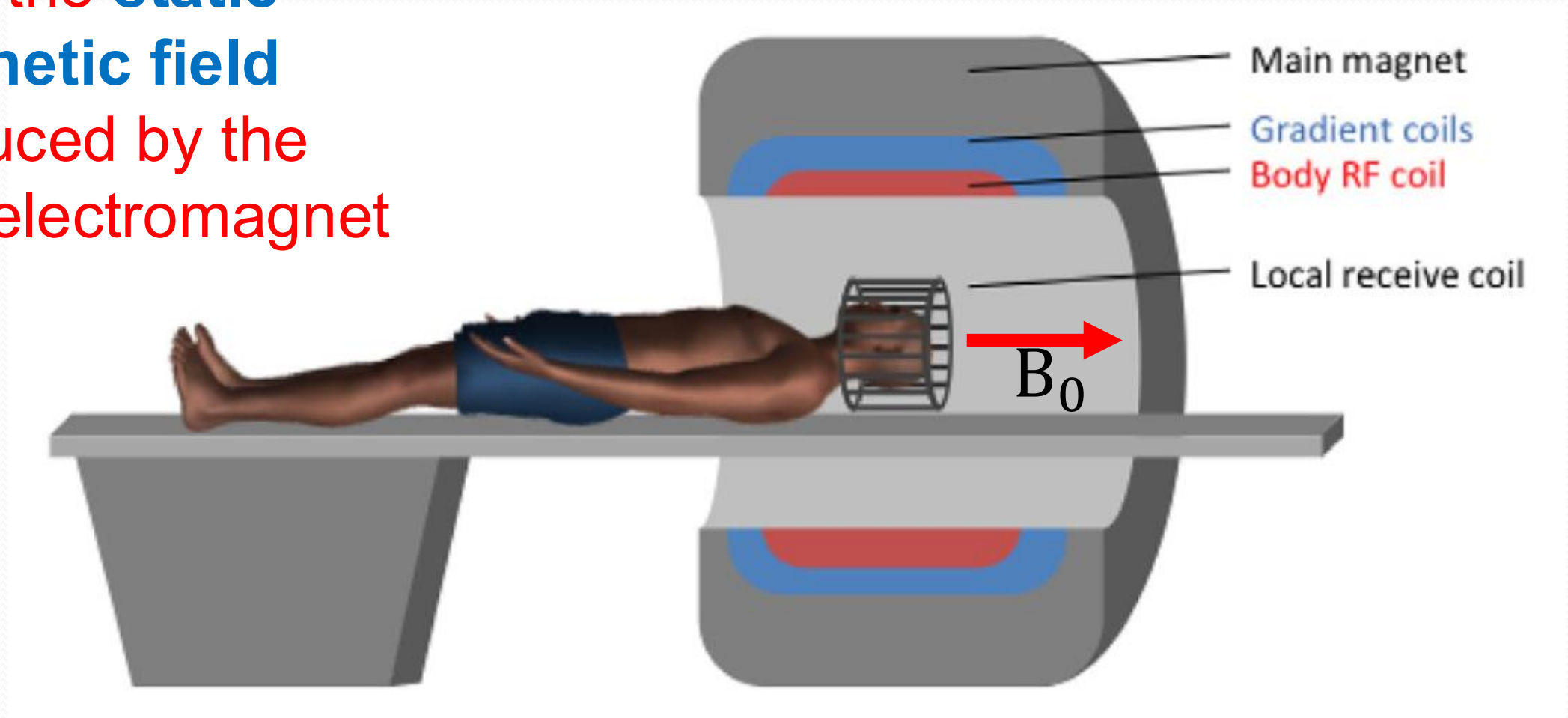
Electromagnet. Solenoid

A **solenoid** is a thin wire wound into a **coil** that produces a magnetic field when an **electric current** is passed through it

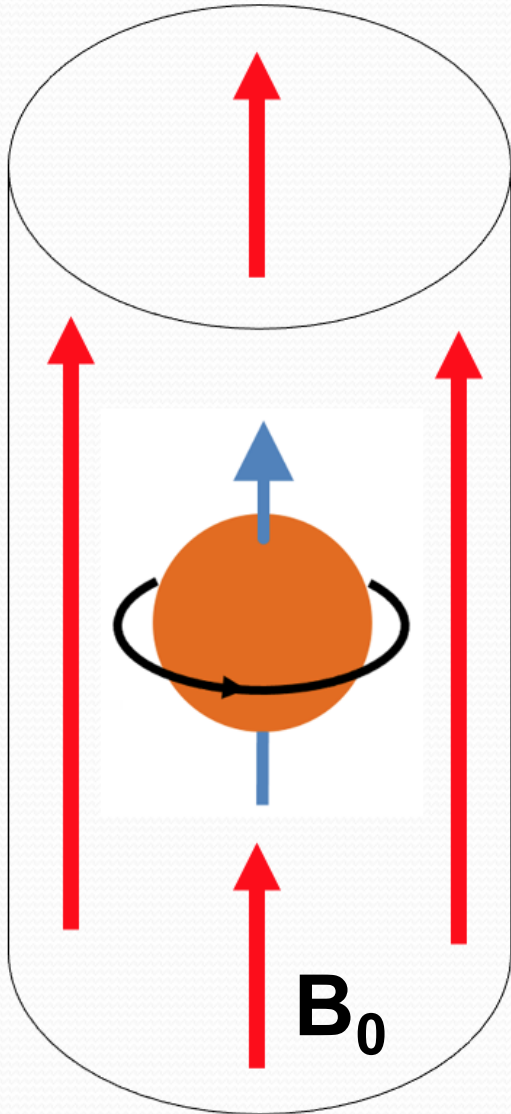


MRI Scanner: Electromagnet

B_0 is the static magnetic field produced by the MRI electromagnet



Magnetic Moment of a Proton

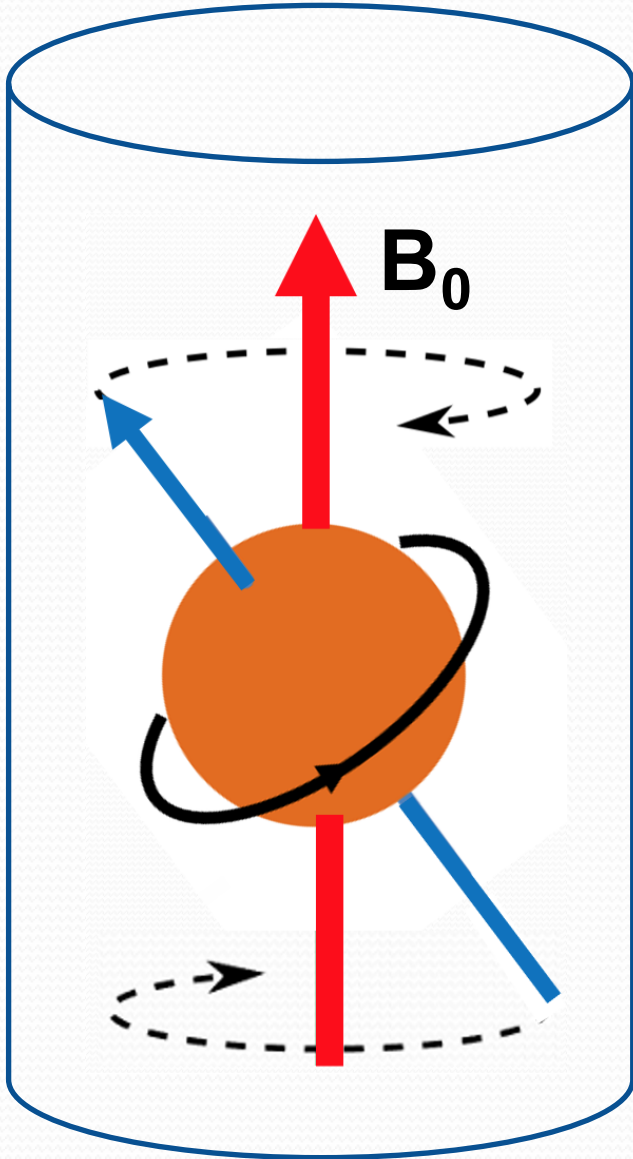


When placed in a magnetic field, a proton experiences a **twisting force (torque)**

The magnitude of this twisting force is determined by the magnitude of the applied magnetic field and the magnitude of **the magnetic moment** of a proton

The value of this **magnetic moment** is an intrinsic property of the proton like its charge and mass

Precession of the Magnetic Moment



The **torque** (a twisting force) due to static magnetic field B_0 causes the following effects:

- **inclination** of the proton's magnetic moment axis relative to the direction of B_0
- rotation of the **inclined** magnetic dipole axis about the direction of B_0 so that it sweeps a conical shape
- This rotation of the magnetic dipole axis is called ***Larmor precession***

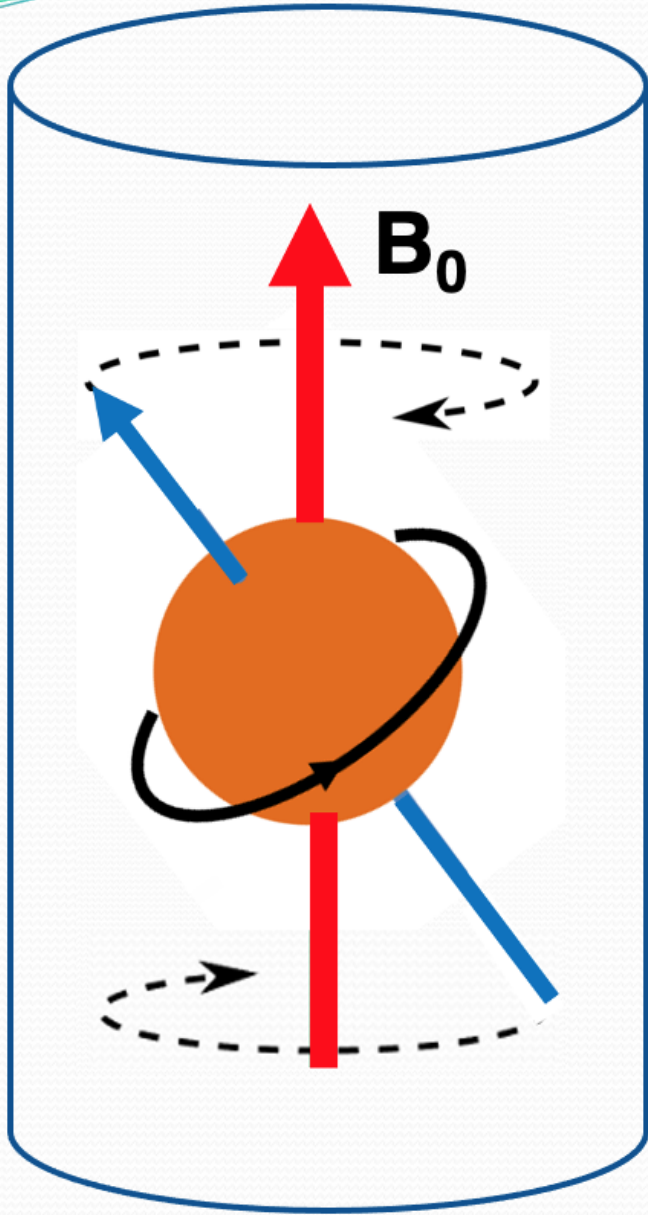
Larmor Frequency of Precession

The number of cycles per second

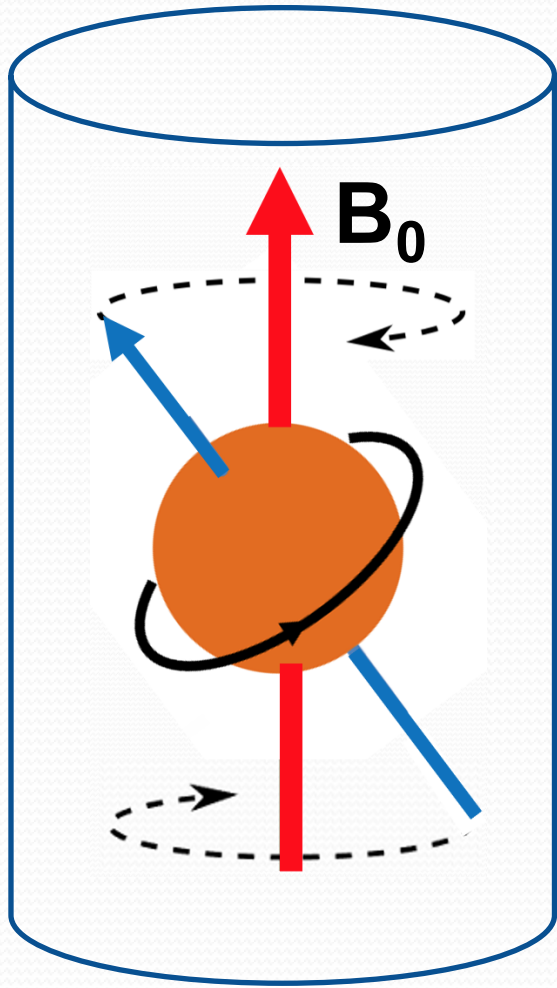
$$f_L = \frac{\gamma}{2\pi} B_0$$

$$\frac{\gamma}{2\pi} = 42.58 \frac{\text{MHz}}{\text{Tesla}}$$

γ is the proton's **gyromagnetic ratio**



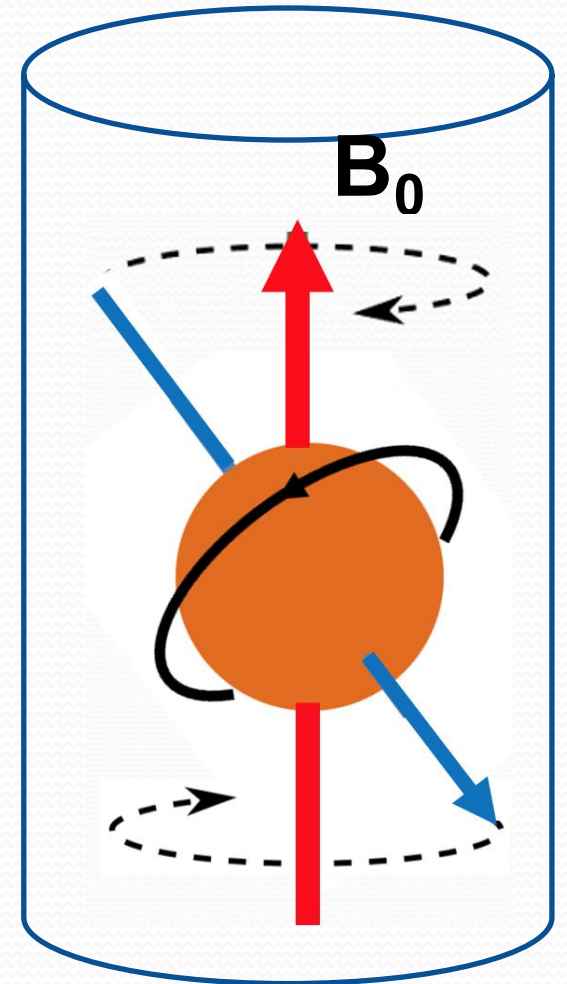
Magnetic Energy of a Magnetic Dipole



Spin-**UP** State

Energy Difference

$$E_{\downarrow} - E_{\uparrow} = h \left(\frac{\gamma B_0}{2\pi} \right)$$



Spin-**Down** State

Nuclear Magnetic Resonance

- The transition of protons from the **low-energy spin-up** state to **high-energy spin-down** state following a photon absorption from a **radio-frequency electromagnetic pulse**
- This transition occurs only if the frequency of the electromagnetic pulse is equal to the proton's **Larmor frequency**:

$$f_{\text{resonance}} = f_L = \frac{\gamma}{2\pi} B_0$$

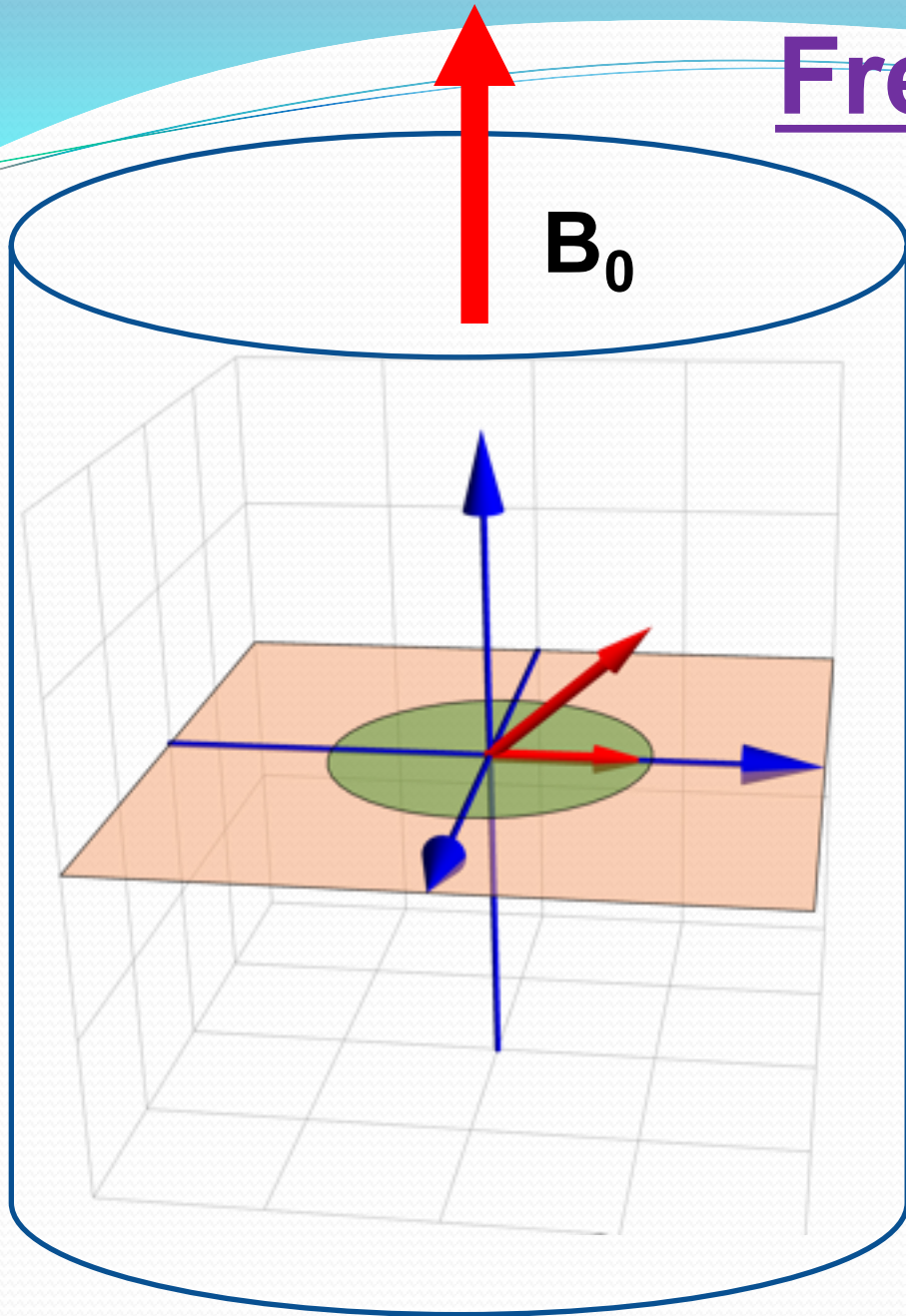
Excitation - Resonance

The **transmitter coil** in an MRI scanner is used to send a radio-wave (RF) pulse with its magnetic component oscillating in the transverse direction

To cause the resonance, the frequency of the RF-pulse has to be **tuned** to the value of the **Larmor frequency**:

$$f_{\text{rf}} = f_L$$

Free Induction Decay Signal (FID)



After the **RF excitation pulse** is applied, the magnetization vector precesses in the **transverse plane** with frequency f_L

Faraday electromagnetic induction:

Precessing magnetization vector induces an oscillating voltage in the receiver solenoid

Free Induction Decay Signal (FID)

- Due to the **Faraday's electromagnetic induction**, the precessing magnetization induces an **oscillating voltage** in the **receiver coil** of the MRI scanner
- This oscillating voltage is called the **free induction decay signal (FID)**

Relaxation of the FID Signal

The exponential decrease of the amplitude of the **FID signal** is caused by two types of processes effecting the magnetic moments of protons:

- **Spin-lattice relaxation**
- **Spin-spin relaxation**

Tissue Contrast in MRI

Tissue contrast in MRI is based on detecting the differences in the **relaxation** of free induction decay (FID) signals

The areas of the tissue emitting a **large amplitude FID signal** correspond to **bright areas** on the MRI image

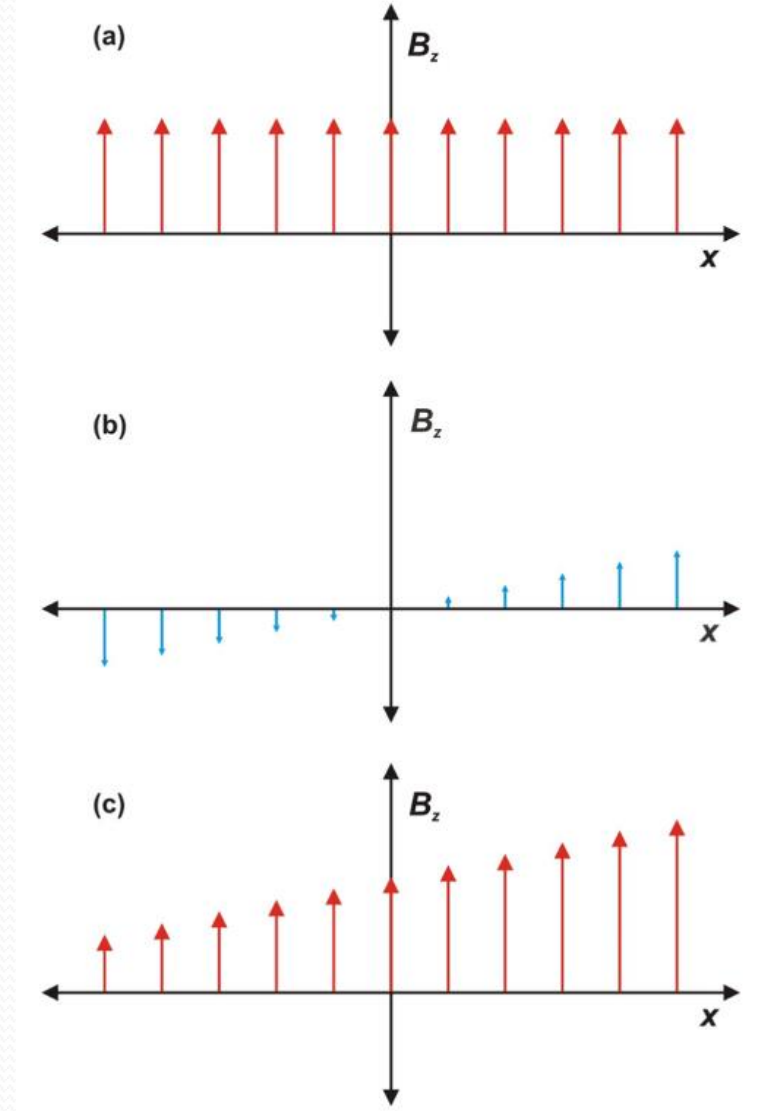
The areas of the tissue emitting a **small amplitude FID signal** correspond to **dark areas** on the MRI image

Spatial localization: Gradient Magnetic Field

Spatial localization in MRI is established using **three gradient magnetic fields** in longitudinal and transverse directions

Gradient field $B_x = Gx$

$$f_L = \frac{\gamma_p}{2\pi} (B_0 + B_x) = \frac{\gamma_p}{2\pi} (B_0 + Gx)$$



Gradient Coils in MRI Scanner

